Environment: Creation Tools

link

This PDF gives you info about how to use the tools included in the asset to help you create your own environment.

Table of contents:

Prefabs

Prefabs Showroom

To discover all prefabs used to create environment contained in this asset:
-open scenes Showroom

Assets > Scenes > Showroom > Showroom

Prefabs folder

Prefabs used to create environment are in the folder Prefabs

Assets > Environment > Prefabs

LODs

In the asset the prefabs that use LODs have the prefix **"LOD_"** at the beginning of their name.

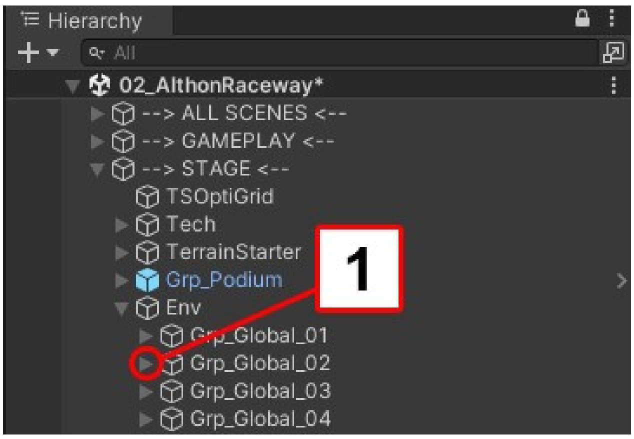
Info:
LOD means “Level of Details”.
Prefabs with different levels of detail included in a LOD appear or disappear depending on the distance from the camera.

Open / close all elements in group in one click:

In hierarchy tab if you want to open all the elements included into a group in one click:

In hierarchy tab :
Hold Shift(Maj) + Left Ctrl + Alt click on the triangle icon near the group (spot 1)

Do the same thing to close all the elements included into a group in one click.



Create group of objects (script)

A script is included in the asset to easily create group:

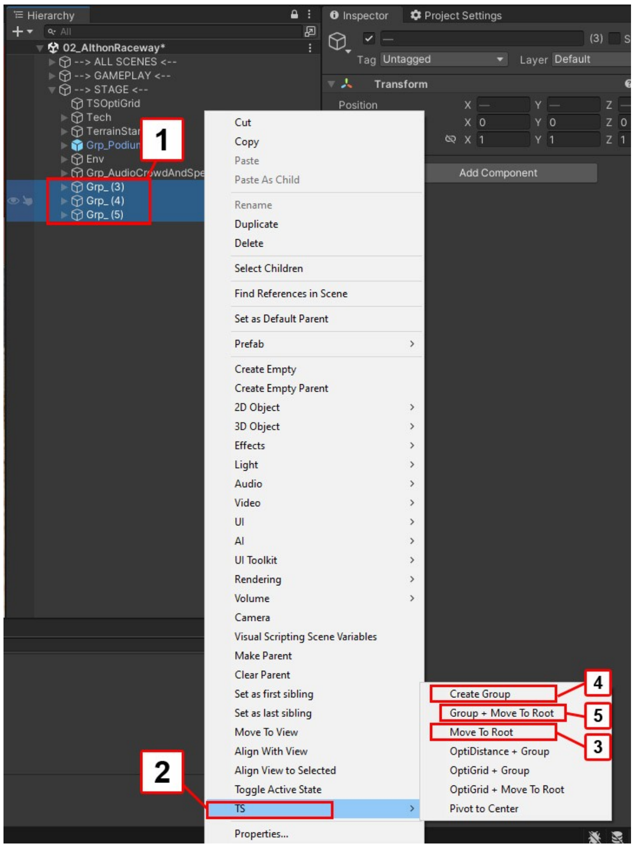
<u>To create a group:</u>
-In hierarchy tab select more than one elements (spot 1)
-Mouse Right click
-In the menu choose TS (spot 2) then Create Group (spot 4)

If you want to create group and move to root of the scene choose create Group + Move to Root (spot 5)

If you want to move select elements to root without making a group choose Move to Root (spot 3)

Important:
The other choice in list:
Optigrid + Group and Optigrid + Move to root are using for optimization.

For learn more information about optimization Link



Procedural objects

Roads, crash barrier, curb and barrier are not prefabs
This elements are created with a script included in the asset.

To learn how to generate road and procedurals features
Link

Flickering

If objects flickering it's probably because two similar objects exactly at the same position.
To solve the issue move one of the two objects a little bit.

Prefabs in details

Object with Physics

Some prefabs add Rigid Body. These prefabs react when cars collide with them.

These prefabs are in folder named Destructible

Assets > Environment> Prefabs > Destructible

Decals

Use decals for ground markings as well as to add ground and terrain details.
The decals are in the folder Decals

Assets > Environment > Decals > Prefabs

If you want more informations about Decals Link

Terrain

Overview

This chapter explain how to create your own map.

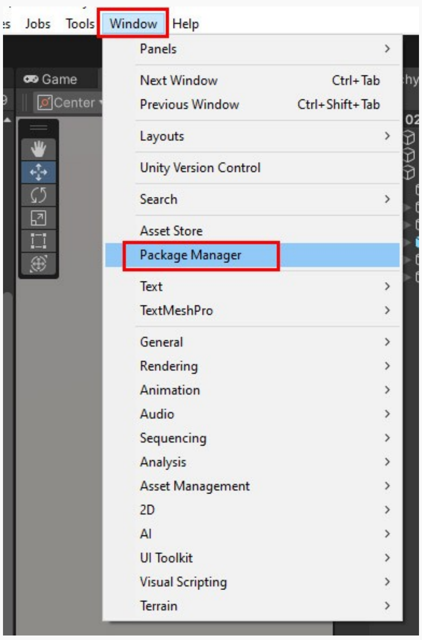
This chapter covers in detail how to:
- Create terrains
- Setup terrains
- Paint textures on terrain

Unity Terrain Tools

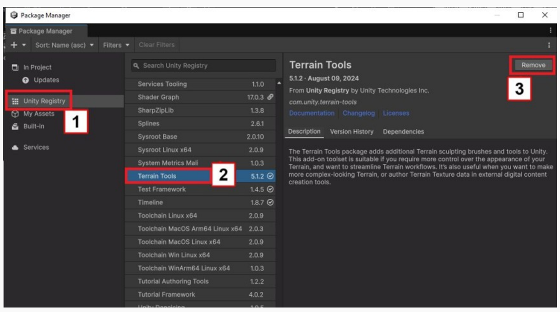
First of all, if you haven't installed the Unity Terrain Tools package, we advise you to do so.
This Unity package is free.
It makes it easier and faster to work on terrain.
The explanations in this chapter will be made using this package.

<u>Here is the process to install the package:</u>

Go to Window > Package Manager



5 -Choose Unity Registry (spot 1)
-Choose Terrains Tools (spot 2)
-Press Install button (spot 3)



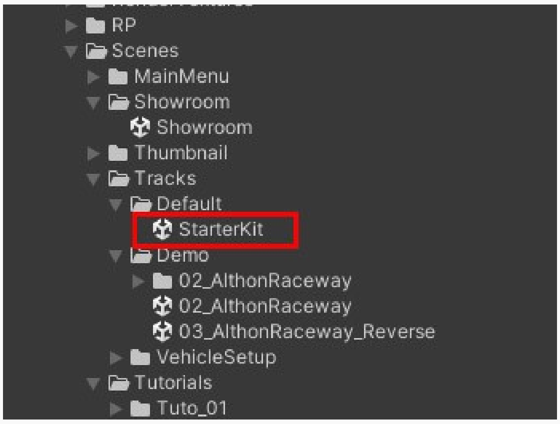
Use Environment Starter kit scene

All the elements to start a new track are included in the starter kit.

Assets > Scenes > Tracks > Default > StarterKit

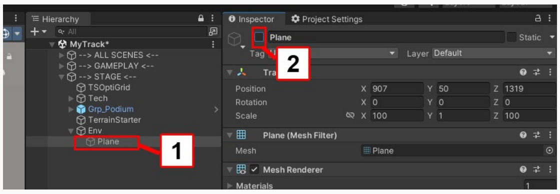
<u>In projet tab:</u>

-Duplicate StarterKit scene
-Rename the copy of the scene for example: MyTrack
-Open MyTrack scene



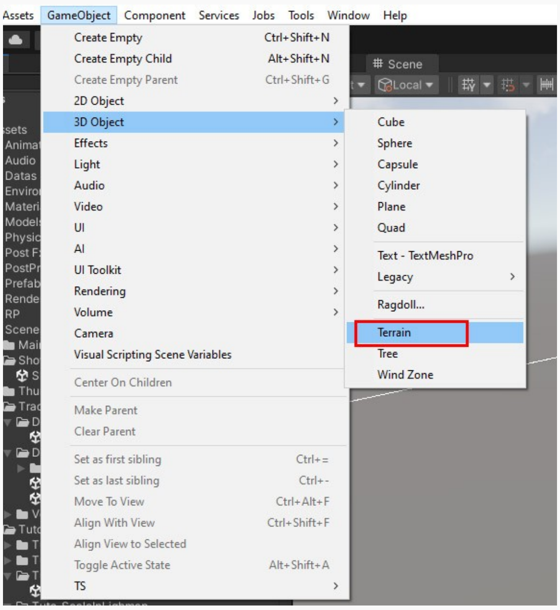
-In hierarchy tab select Plane (spot 1)
STAGE > Env > Plane

-In Inspector tab disable Plane (spot 2)



-Create a new Terrain:
GameObject > 3D Object > Terrain

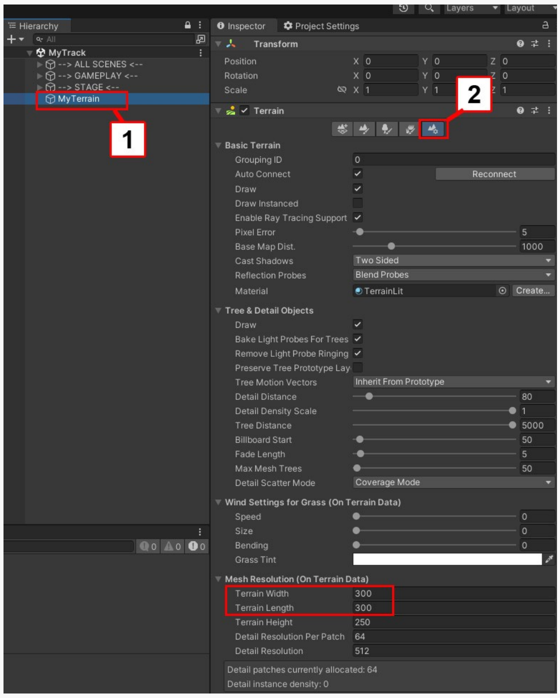
-Rename the new Terrain for example: MyTerrain



In Hierarchy tab:
select MyTerrain (spot 1)

In Inspector tab:
- Press Terrain Settings button (spot 2)

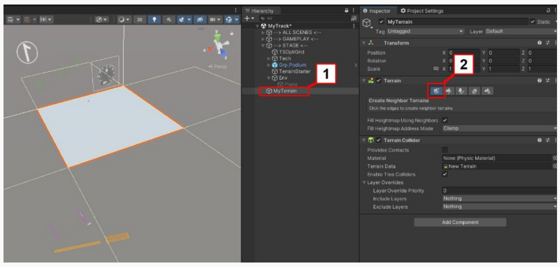
-Set Terrain Width to 300
-Set Terrain Length to 300



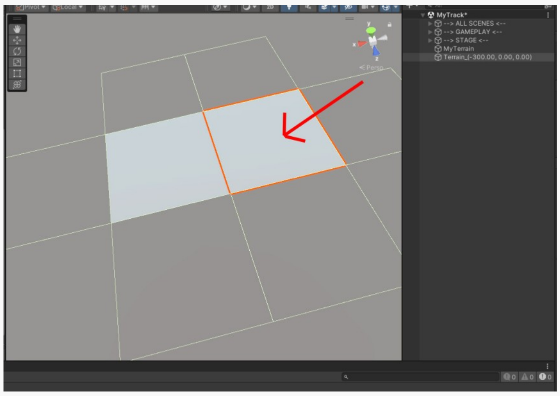
Add terrain neighborhood

In Hierarchy tab:
select MyTerrain (spot 1)

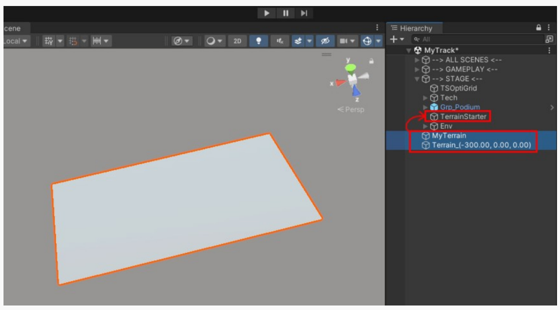
In Inspector tab:
- Press Create Neighbor Terrains button (spot 2)



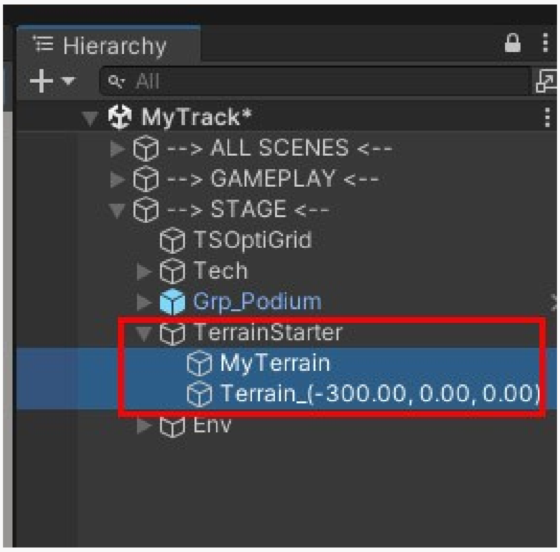
In Scene tab:
- Click on the square to create a new terrain



In Hierarchy tab:
- Add the 2 terrains into TerrainStarter



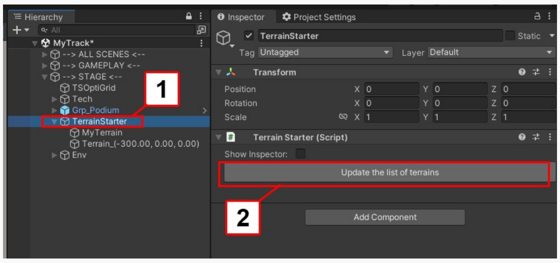
You must have this result:



Update list of terrains

In Hierarchy tab:
-Select TerrainStarter (spot 1)
STAGE > Terrain Starter

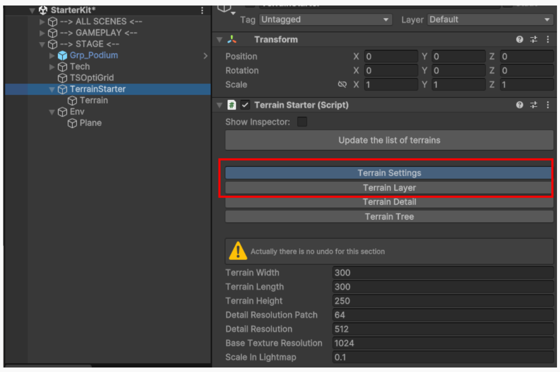
In Inspector tab:
-Press Update the list of terrains (spot 2)



In Hierarchy tab:
- Select TerrainStarter
STAGE > Terrain Starter

On TerrainStarter there is a script named TerrainStarter.cs
This script allows to configure terrains in a few clicks.

This script allows to setup:
- terrains parameters
- terrains textures layers



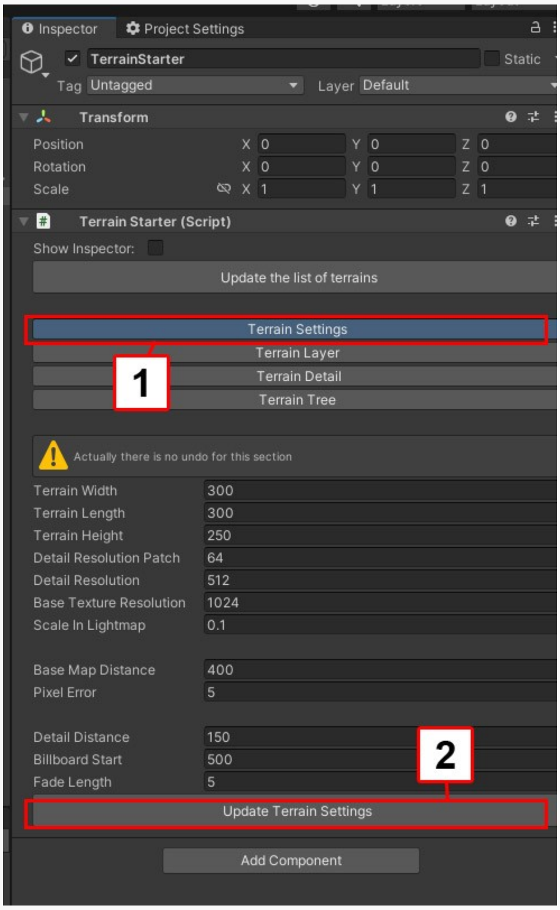
Very important:
Each time you add or remove terrain into TerrainStarter you must press Update the list of terrains button.

Setup terrain

In hierarchy tab:
-select TerrainStarter

In Inspector tab:
- Press Terrain Settings button (spot 1).
- Press Update Terrain Settings button (spot 2).

Parameters of the terrains are modified

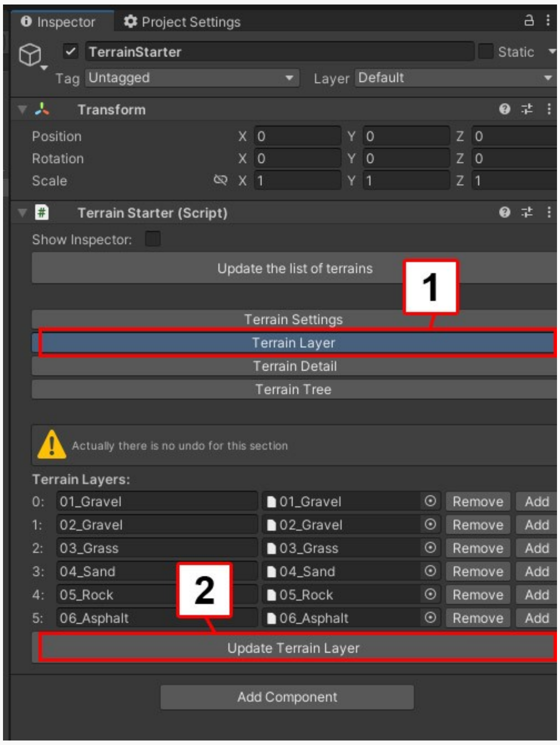


Important:
All the terrains that are in group TerrainStarter will be modified by the script

In hierarchy tab:
-select TerrainStarter

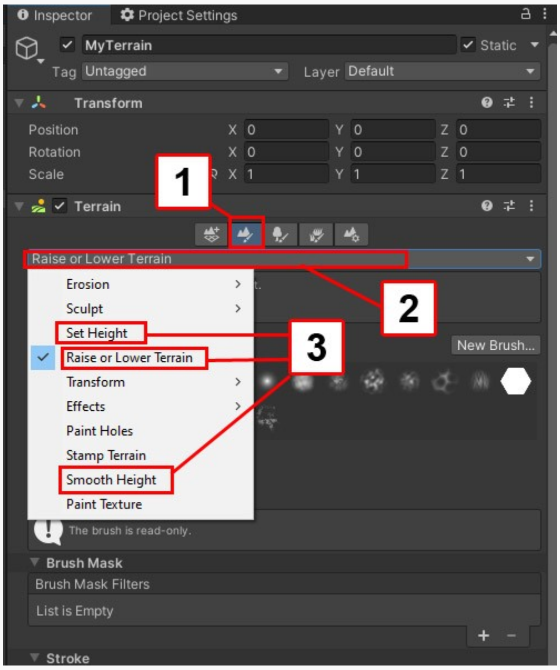
In Inspector tab:
-Press Terrain Layer button (spot 1).
-Press Update Terrain Layer button (spot 2).

Layers Textures are added to all the terrains included into TerrainStarter



Tips for Terrain sculpting

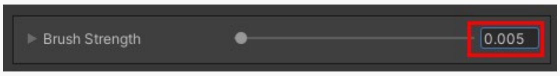
<u>To sculpt the terrain:</u>
-In hierachy tab select a terrain
-In Inspector tab select the second icon (spot 1).
-Click on the menu selection bar (spot 2).
-Choose Raise or Lower Terrain, Set Height or SmoothHeight (spot 3).



Learning how to use terrain sculpt tool is beyond the scope of this documentation but here some tips to help you get good result.

<u>Raise or Lower Terrain:</u>

For small details use low Brush Strength value (for example 0.005 to 0.01)

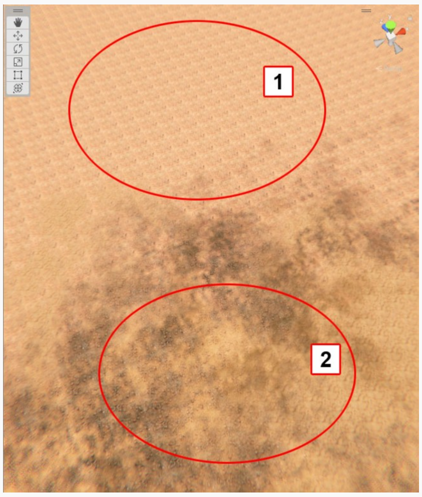


<u>SmoothHeight</u>

To smooth harder increase the Blur Radius value (for example 10)
Work with large brush size

Custom brush

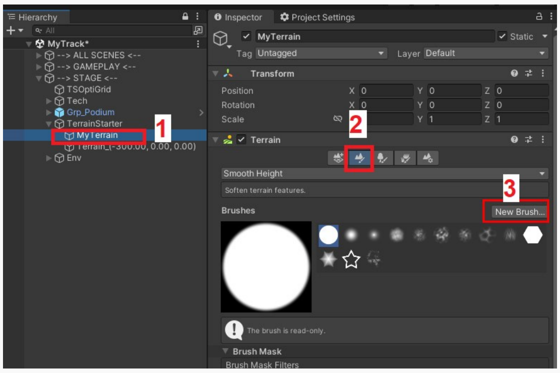
A custom brush is included in the asset.
This custom brush is designed to reduce texture tile effect.



In the picture on the right:

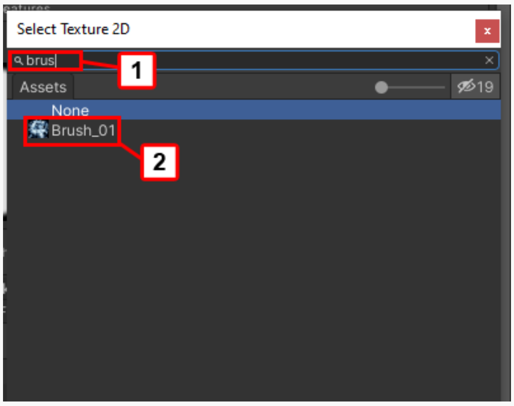
Texture tile is visible (spot 1).
Texture tile is less visible (spot 2).

To install the brush:



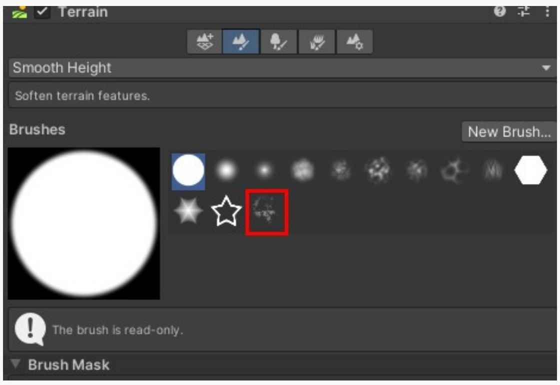
If Brush_01 is not automatically installed:

-Select a terrain (Spot 1)
- In the Inspector select Paint Terrain icon (spot 2)
- Press New Brush button (spot 3)



-In Text Field type “brush” (spot 1).
-Double click on Brush_01 (spot 2).

A new brush is added in brushes Panel



Paint terrain texture

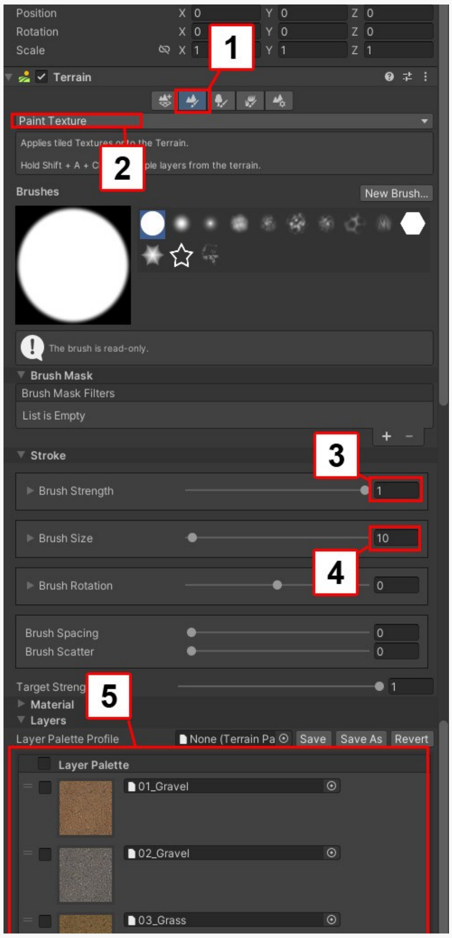
<u>To paint a terrain:</u>
-In hierachy tab select a terrain

-In Inspector tab select Paint Terrain icon (spot 1).
-Click on the menu selection bar
Choose Paint Texture (spot 2).

-Set Brush Strenght to 1 (spot 3).

-Set Brush Size to 10 (spot 4).

-Select a texture to paint (spot 5)



Roads and Procedural features

Roads

Overview

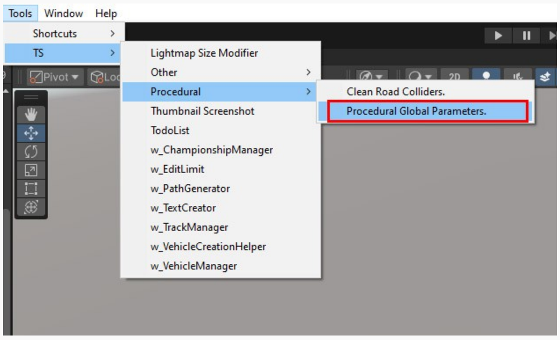
A script allowing to create procedural elements is included in the asset

<u>Script allows to create:</u>
-Roads (adapts to the terrain)
-Curb
-Crash barrier
-Road block
-Wall + metal fence
-Wood Barrier
-Special Collider

Important :
Objects created with the script are not prefabs. Added this objects increase the size of scenes more than prefabs.
It is better not to copy/paste these objects.

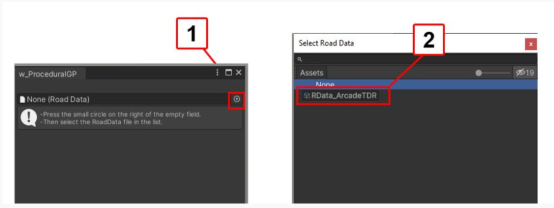
Setup

Go to Tools > TS > Procedural > Procedural Global Parameters



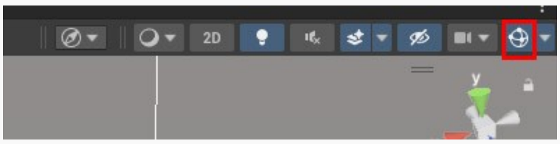
-Click on circle icon (spot 1)

-Choose Rdata_ArcadeTDR (spot 2)



Info:
This setup should only be done the first time the panel is opened

Important:
In scene view:
-click on gizmos icon to activate gizmo



Create roads

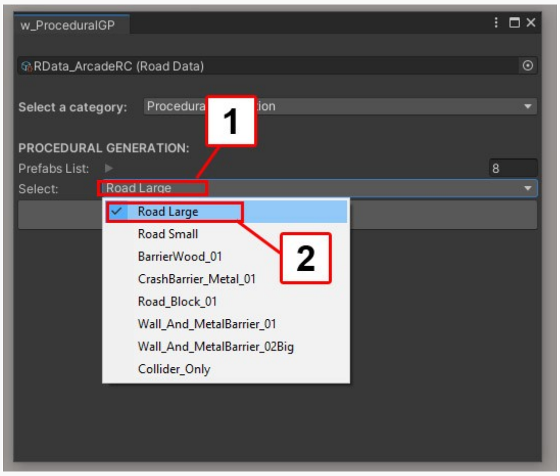
In Project tab double click on Tuto_Procedurals_01 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

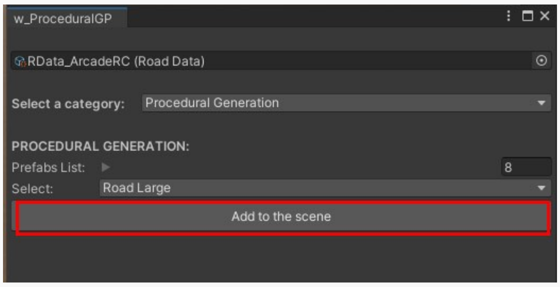
Into the procedural panel:

-Click on select bar (spot 1)

-Choose Road Large (spot 2)

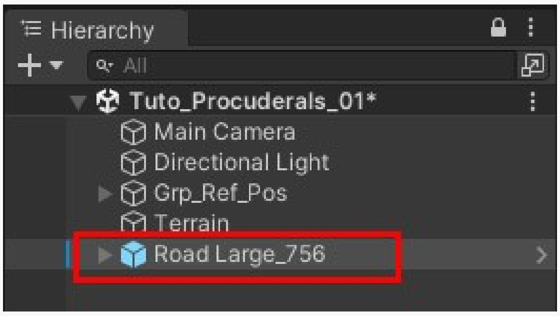


-Press Add to the scene

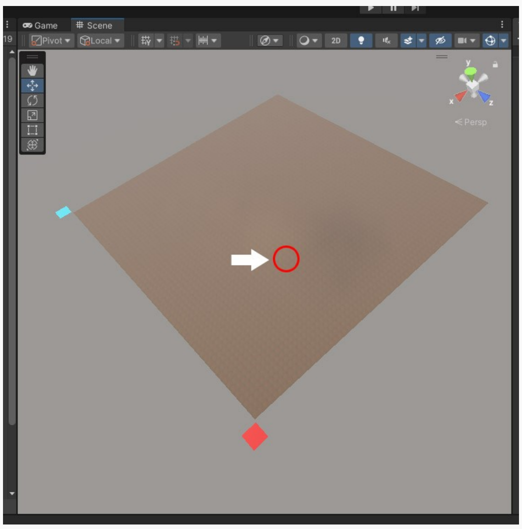


In hierarchy tab:
A new road is created.

-Select the new road



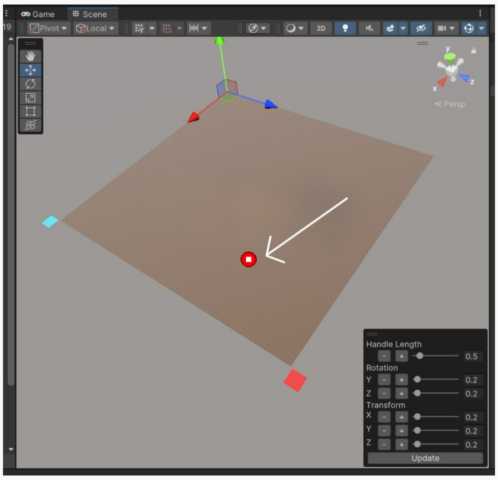
-Position the mouse over the terrain where you want to start the road



Into scene view:
To be sure the scene view is focus:
-select Hand Tool

Press keyboard shortcut N

The first point is created



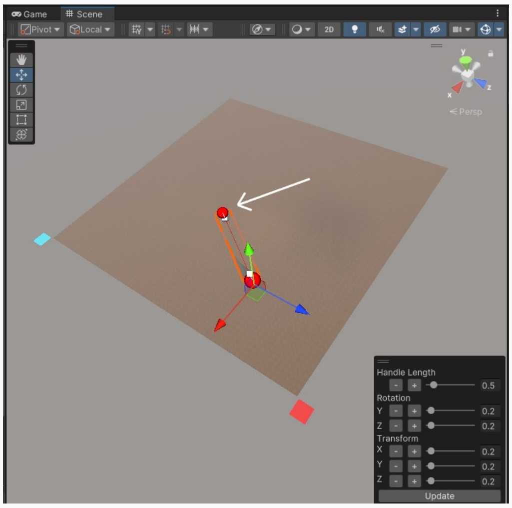
Important:
If the first point doesn't appear:

-Verify that gizmos are activate in scene tab
-In scene view select the terrain
-In hierarchy tab select again the new Road
-Press keyboard shortcut N

-Position the mouse over the terrain where you want to create the second point.

Don't forget:
To create a point, the road **must be selected** in the hierarchy tab.

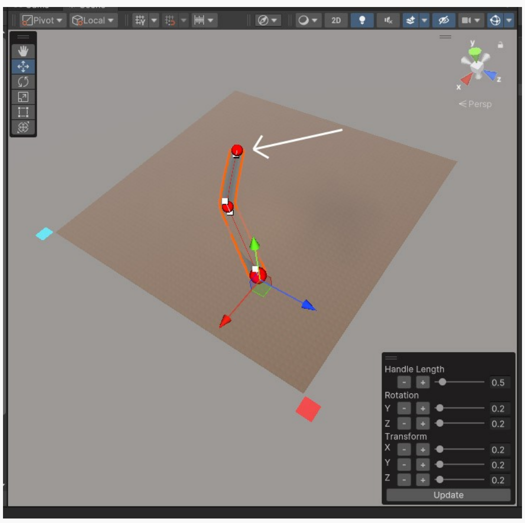
-Press keyboard shortcut N
The second point is created



Info:
if you want to undo:
-Press keyboard shortcut Ctrl+Z

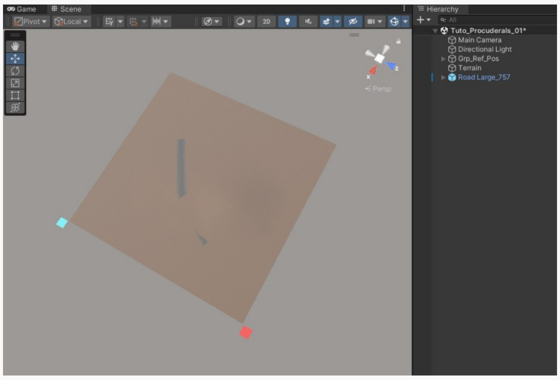
-Position the mouse over the terrain where you want to create the third point.

-Press keyboard shortcut N
The third point is created



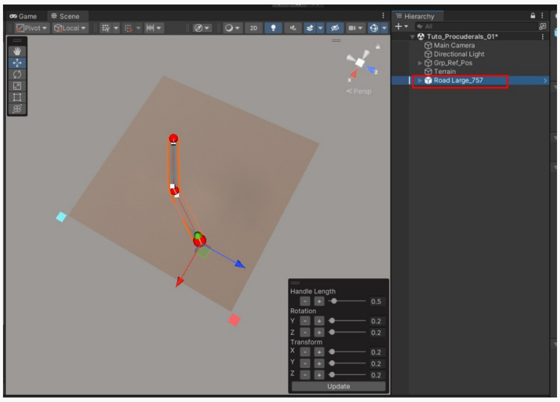
-In hierarchy tab unselect the road

Road points are no longer selectable



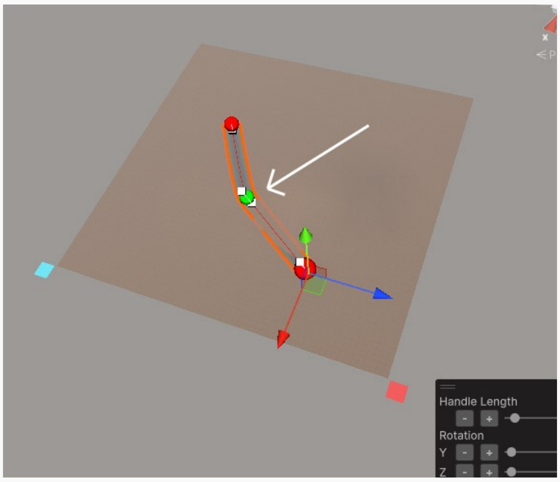
-In hierarchy tab select the road (spot 1)

Road points are selectable (spot 2)

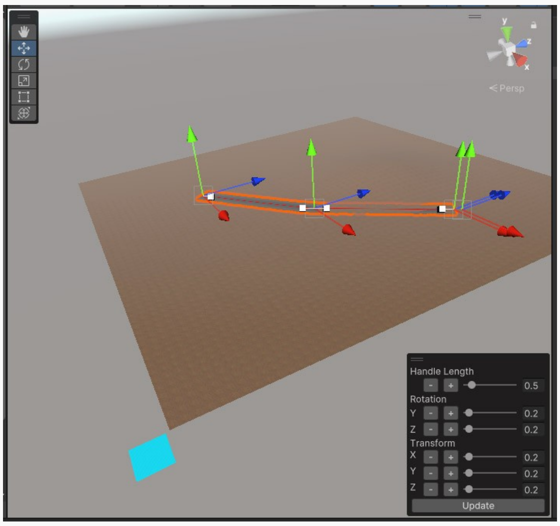


-Click on the second point to select it

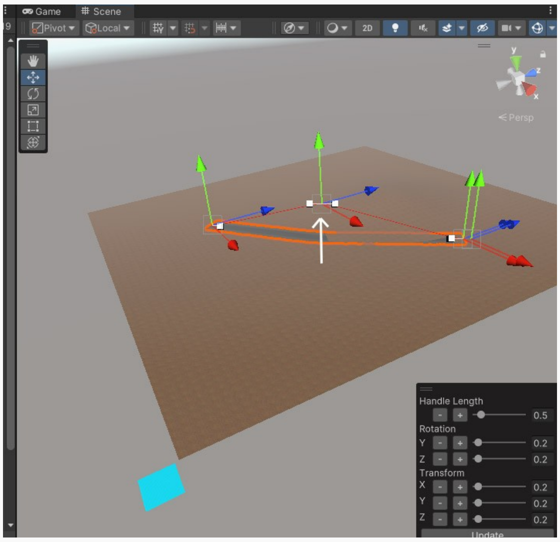
Point turns green



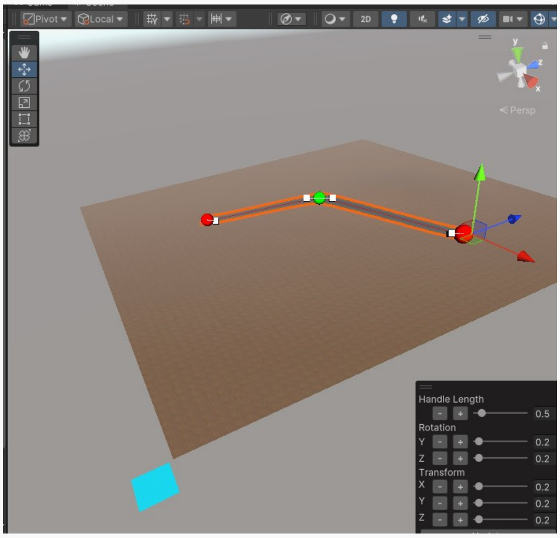
-Hold Shift (Maj)
X Y Z translation handles appear



While holding down the Shift (Maj) move handle on Y axis.

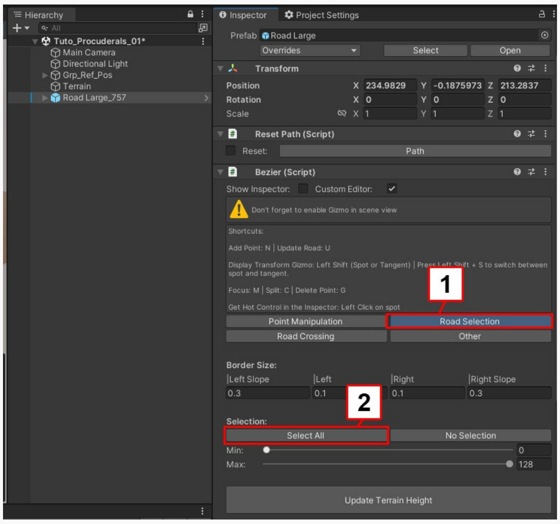


-Press U to update the Road

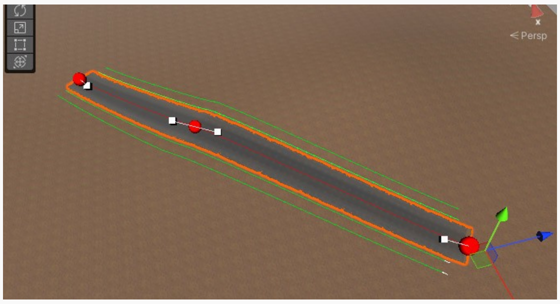


-In Inspector tab
Press Road Selection button (spot 1)

Press Select All button (spot 2)

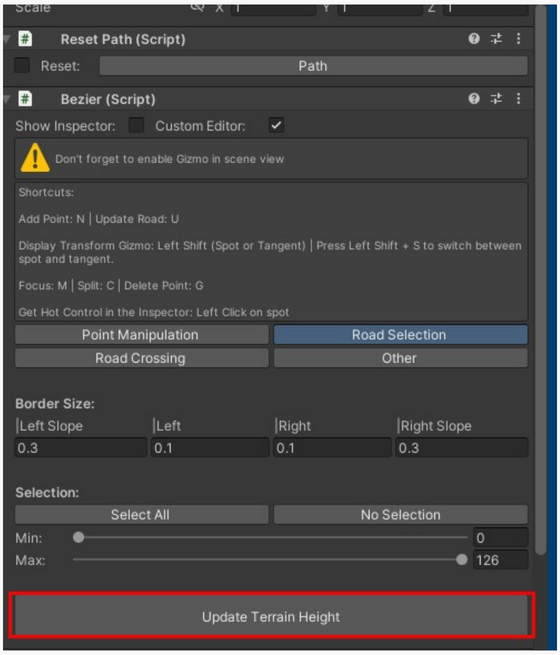


Green lines appear on the sides of the road

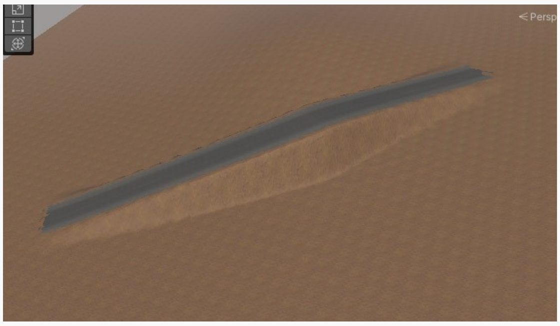


Info:
The green lines represent the distance used to make a smooth transition with the terrain

-In Inspector tab
Press Update Terrain Height button

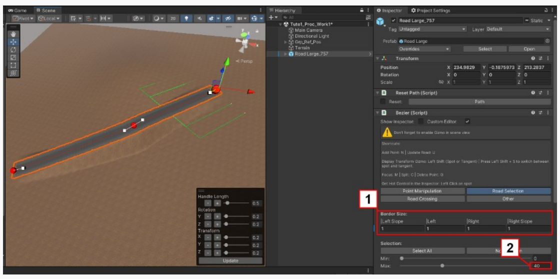


The terrain has adapted to the shape of the road



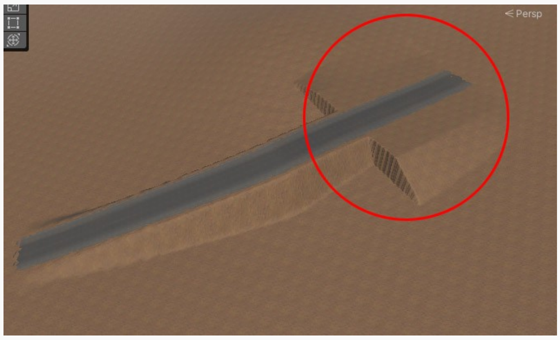
-In Inspector tab
In Border size section: (spot 1)
set Left slope to 1
set Left to 1
set Right to 1
set Right Slope to 1

In Selection section: (spot 2)
set Max to 40 so that only half of the road will be affected



-In Inspector tab
Press Update Terrain Height button

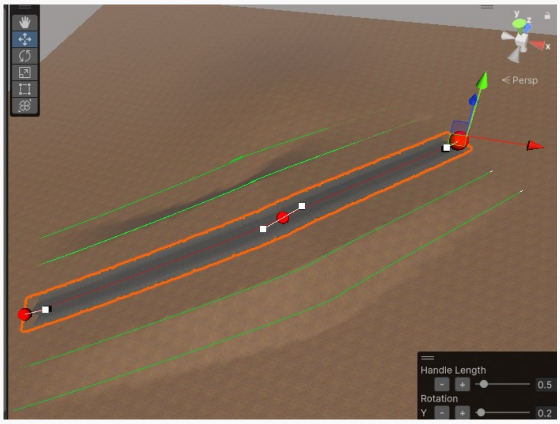
Transition between terrain and road is smoother but only at the beginning of the road



-In Selection section:
Press Select All button (spot 1)

-In Inspector tab
Press Update Terrain Height button (spot 2)

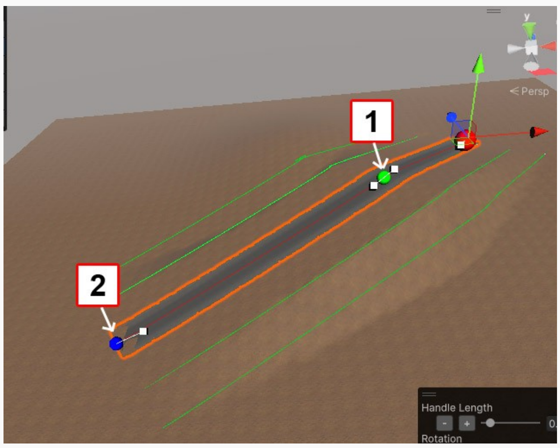
Transition between terrain and road is smoother all along the road



<u>To select several points at the same time</u>

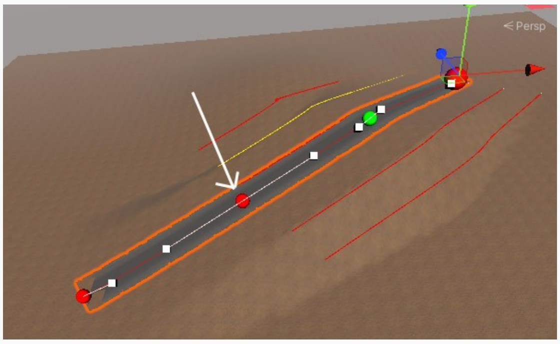
-Select a Point (spot 1)
Point turns green

-While holding down the Left Ctrl select a second point
Second point turns blue (spot 2)



-Select 2 Points
-Press C to split

A new point is created between the 2 points

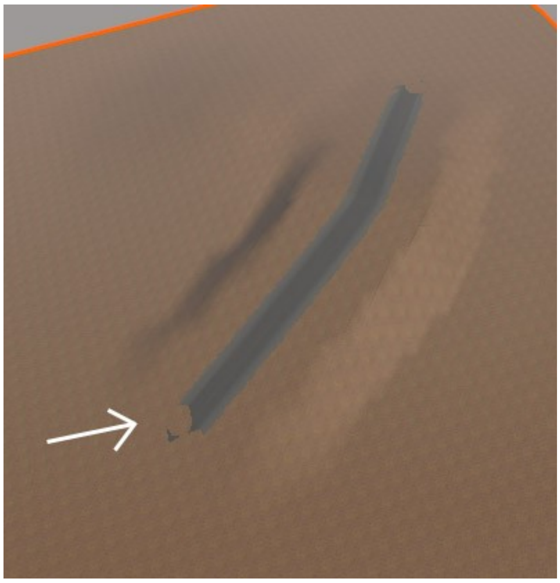


-Select the new point.
-Press G to delete the point

Important:
Don't forget to press U when you want to update the shape of the road

Artefact when the camera is far from the terrain

As the camera move away the definition of the terrain decreases. The terrain no longer fits perfectly with the road.

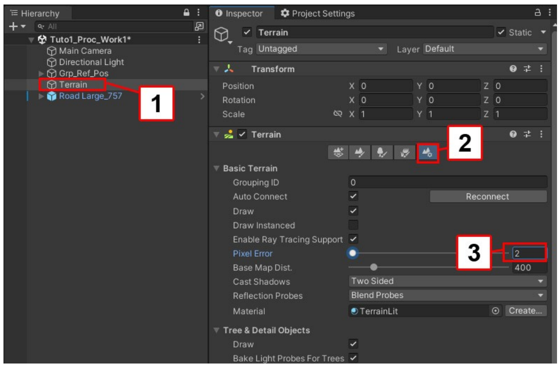


To solve this problem there are 2 solutions:

Solution 1: reduce Pixel error value

-In hierarchy tab select the terrain (spot 1)
-In Inspector tab press button Terrain settings (spot 2)
-Set Pixel Error to 2 (spot 3)

Warning: Lowering the pixel value increases the definition of the terrain. The terrain is therefore less optimized.



Solution 2: Paint asphalt texture on terrain

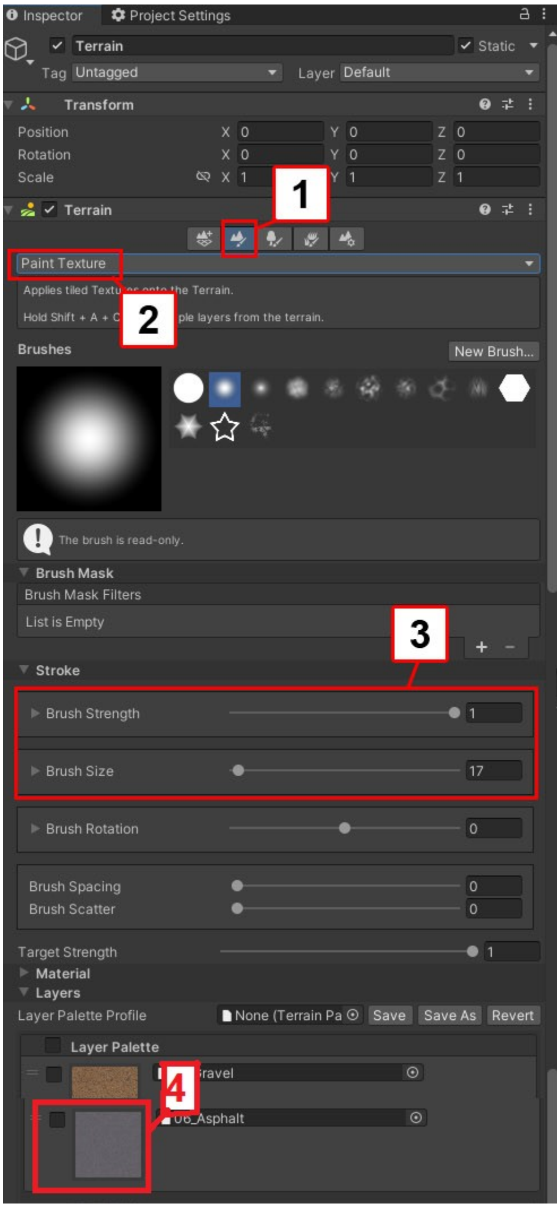
In hierarchy tab:
Select Terrain

In Inspector tab:
-Press button Paint Texture (spot 1)

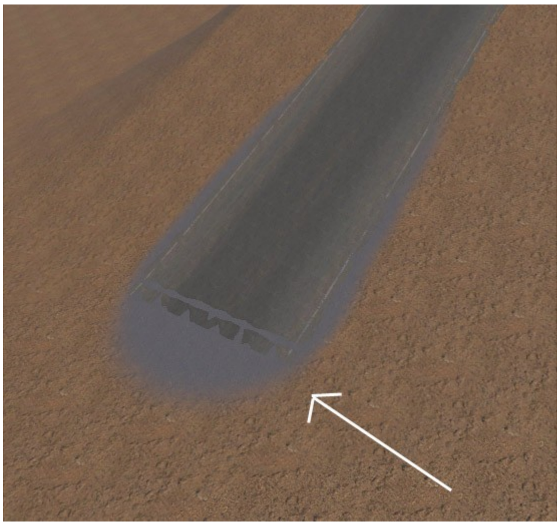
-Choose Paint Texture (spot 2)

-Set Brush Strenght to 1
-Set Brush Brush Size to 17 (spot 3)

-Select 06_Asphalt_Road texture (spot 4)



Paint the asphalt texture under the road

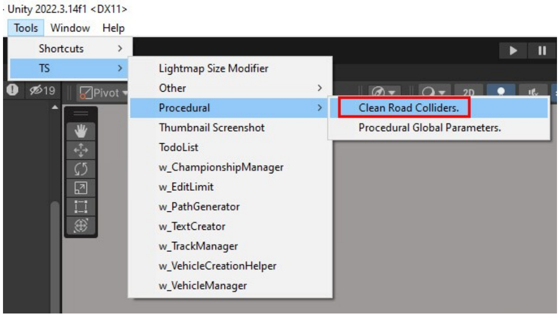


Road colliders

To adapt the terrain to the road, colliders are created.
Once the road is adapted to the terrain these colliders are no longer useful.
It is necessary to delete them.
If you do not remove them, it is possible to have invisible collisions at certain places of the terrain

Go to Tools > TS > Procedural > Clean Road Colliders

All the roads colliders are remove.



Important:
If you modify the roads again, new colliders will be created.
Don't forget to delete them.

Modify the curve of the road

To modify the curves there are 2 solutions
-a tool box
-a keyboard shortcut (Advance mode)

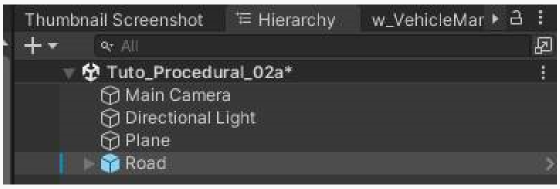
You can choose one or the other solution according to your needs

Solution 1: Ovelay toolbox

In Project tab double click on Tuto_Procedurals_02 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

- In the Hierarchy select Road object



The first time the overlay is not displayed in the scene view:

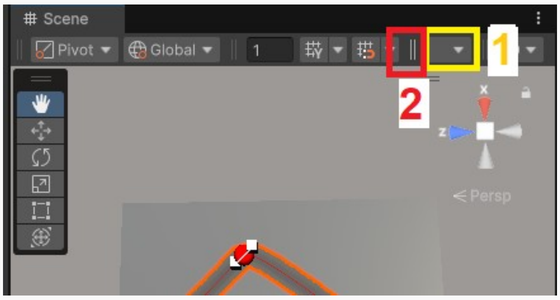
-Don't forget to select Road object

-Find the icon (spot 1) in the scene view bar.

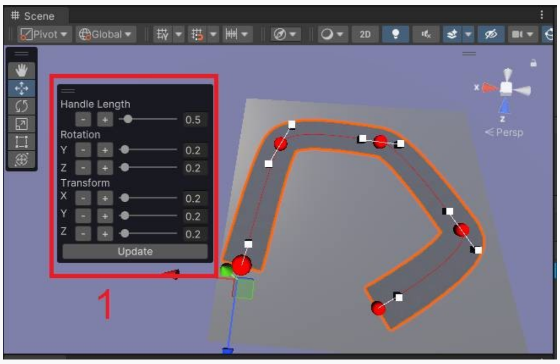
-Then Hold click on || (spot 2)

-Drag and drop the container in the scene view.

-Finally release the left click to see the panel.



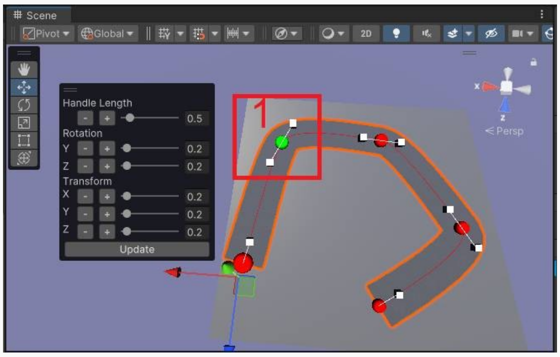
- Now in Scene view the overlay is displayed (spot 1).



This overlay allows you to modify for each red point of the road:
The position of the point
The length of its Handle
The rotation of its Handle

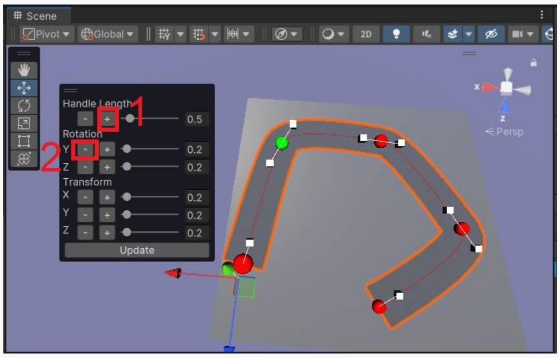
- As an example: Select the second point of the road (spot 1)

The spot become green.

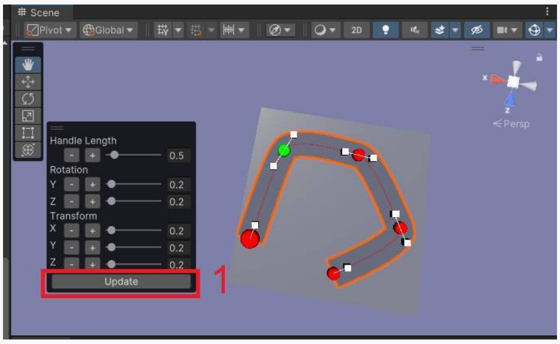


- Press several times Handle Length + button (spot1).
The size of the handle increase.

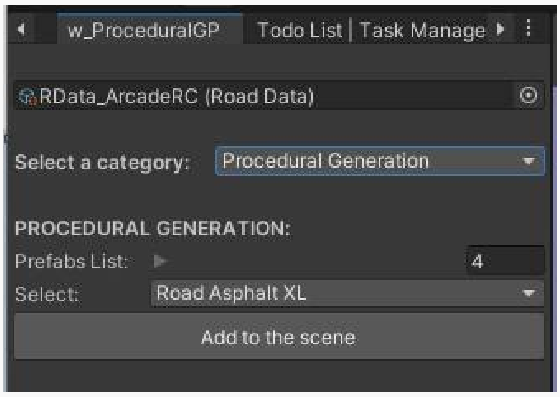
- Press several times Rotation Y – button (spot 2).
Rotation on Y axis is applied.



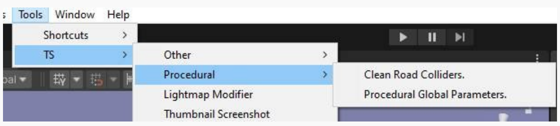
- Press Update button to apply the modifications on the road.



Note:
It is possible to use this process for each path created with the w_ProceduralGP window



Tools > TS > Procedural > Procedural Global Parameters



Note:
For each parameter it is possible to change the strength of the parameter.

Solution 2: keyboard shortcut (Advance mode)

In Project tab double click on Tuto_Procedurals_03 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

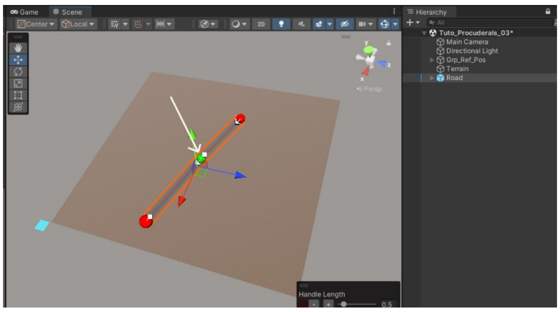
<u>In the **scene** tab:</u>
-select Road

Very Important:

Don't select the road into the hierarchy tab.
This can cause problems when trying to access the curve handles.
Select the road into the scene tab.

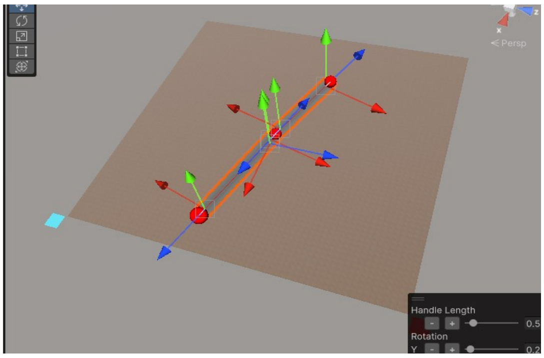
-Select the point of the road as indicated on the image on the right

Point turns green

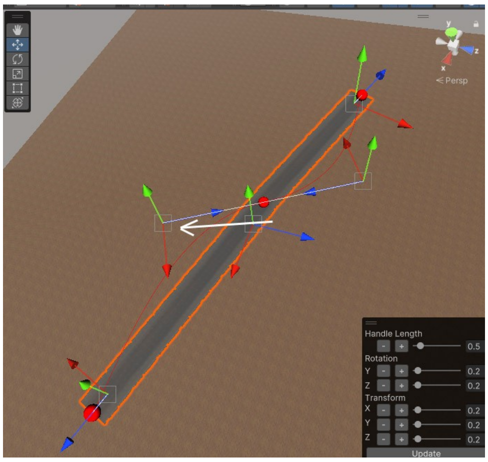


-While holding down the Shift (Maj) press S then release S

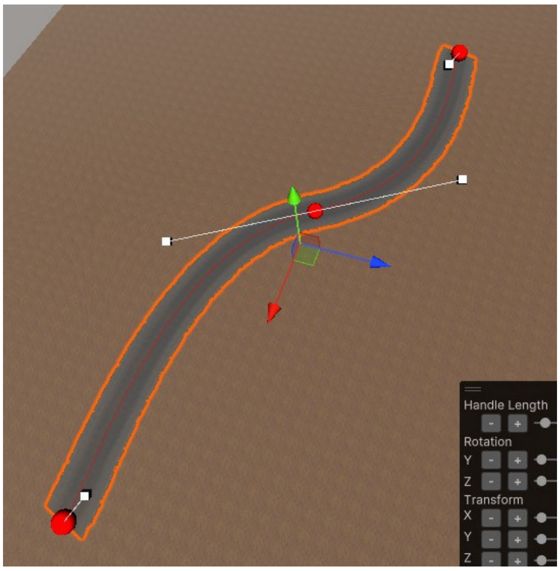
New handle appears



-While continuing to hold down the Shift (Maj), move red axis as indicated on the image on the right



-Press U to update the Road



Align points on curves

In Project tab double click on Tuto_Procedurals_04 scene to open it.

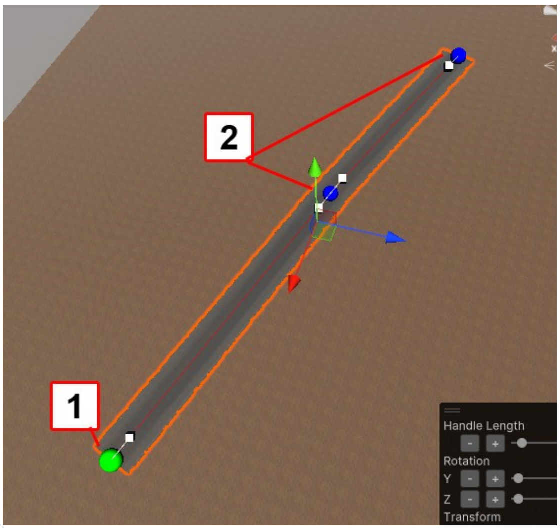
Assets > Scenes > Tutorials> Tuto_Procedurals

-In hierarchy select Road

In scene tab:

-Select the first point of a curve (spot 1)

-While holding down the Left Ctrl select the other points (spot 2)

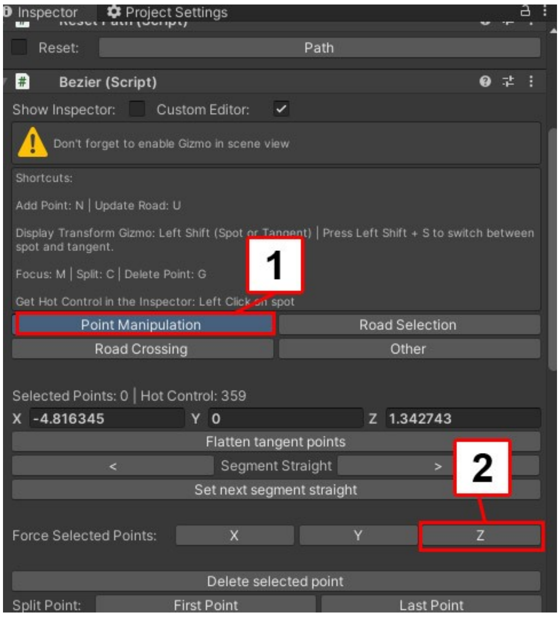


In Inspector tab

Press Point Manipulation button (spot 1)

-Press Z button to align points on Z axis (spot 2)

-Press U to update the Road



Create crossroad

In Project tab double click on Tuto_Procedurals_05 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

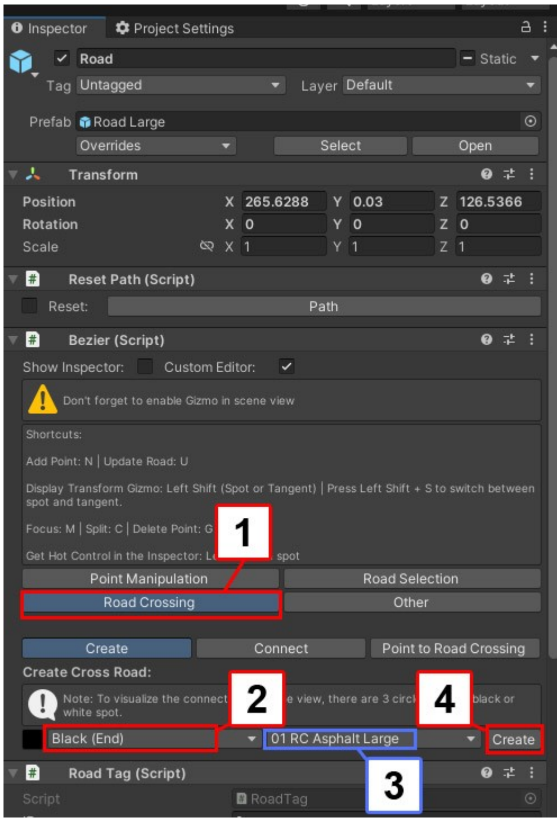
-In hierarchy select Road

-In Inspector tab
Press Road Crossing button (spot 1)

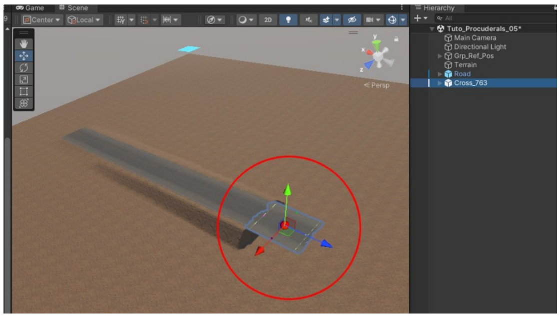
-Choose Black(End) (spot 2)

The type of road is automatically configured to match the road type (spot 3)

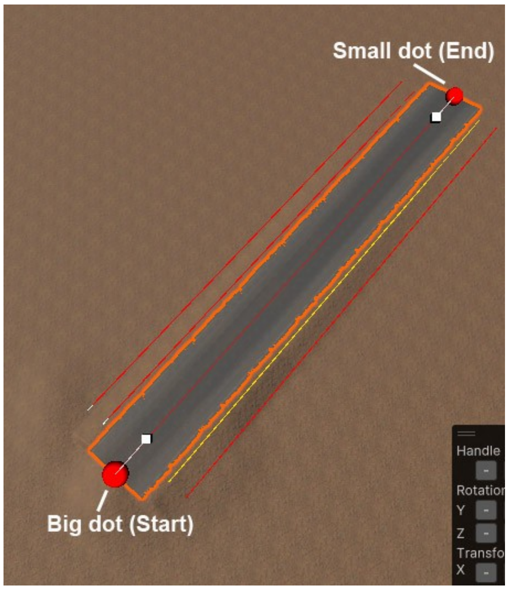
-In Inspector tab
Press Create button (spot 4)



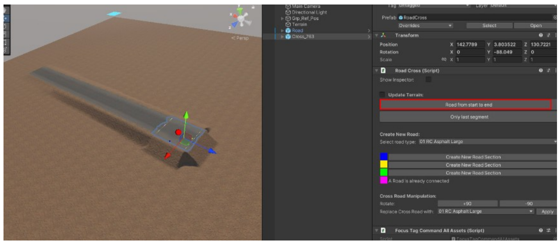
A new crossroad is created



Info:
the dot that indicates the start of the road is bigger than the dot that indicates the end of the road

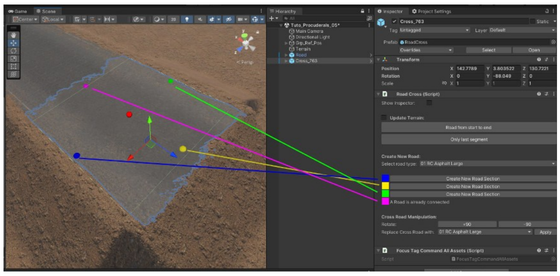


-Press Road From Start to End button to adapt the terrain to the new crossroad

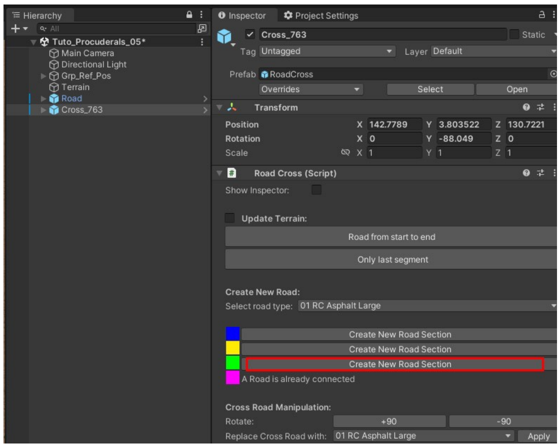


-In hierarchy tab select the crossroad

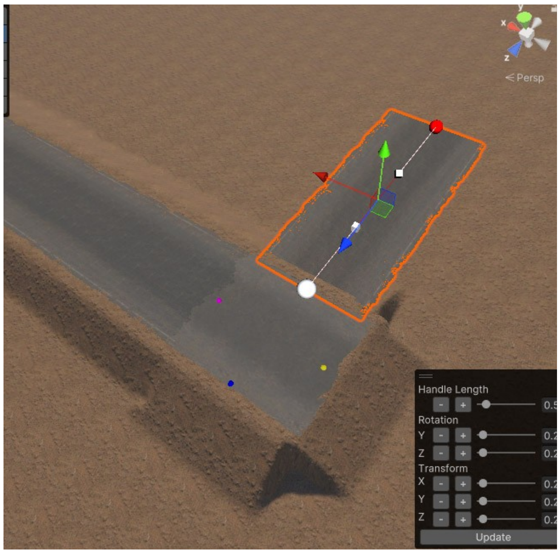
A different color indicates each side of the crossroads



-Press Create new road section in front of the green square



A new road is created from the crossroads

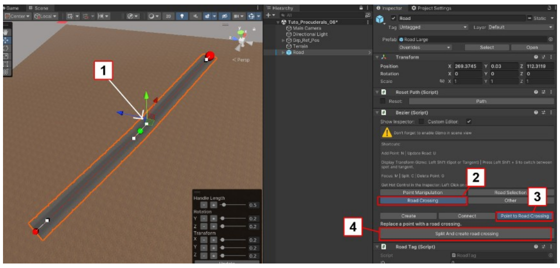


Create a crossroad from a point

In Project tab double click on Tuto_Procedurals_06 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

-In the scene tab:
select Road



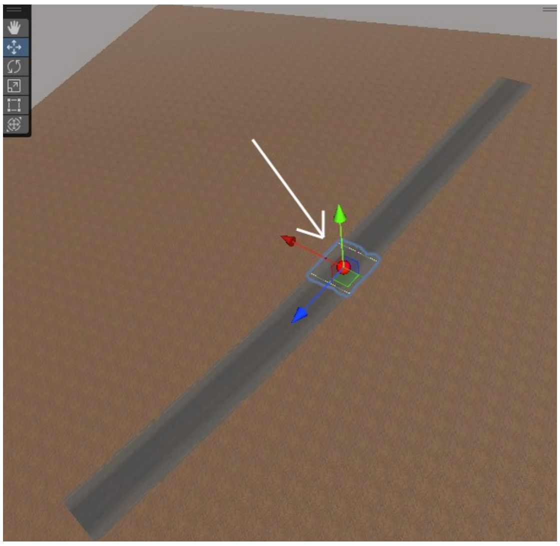
-Select the point in the middle of the road curve (spot 1)
Point turns green

-In Inspector tab:
Press Road Crossing button (spot 2)

-Press Point To Road Crossing button (spot 3)

-Press Split And create road crossing button (spot 4)

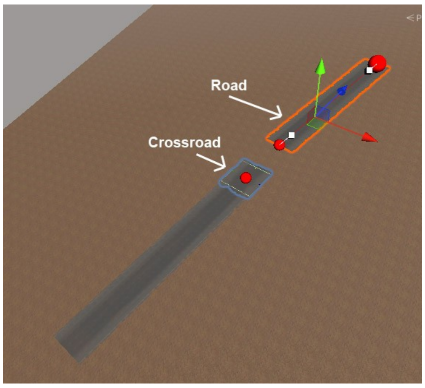
A new crossroad is created



Connect a road with a crossroad

In Project tab double click on Tuto_Procedurals_07 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

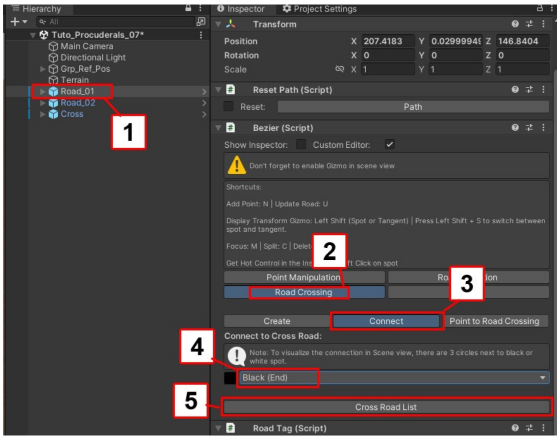


-In hierarchy tab select Road_01 (spot 1)

-In Inspector tab:
Press Road Crossing button (spot 2)

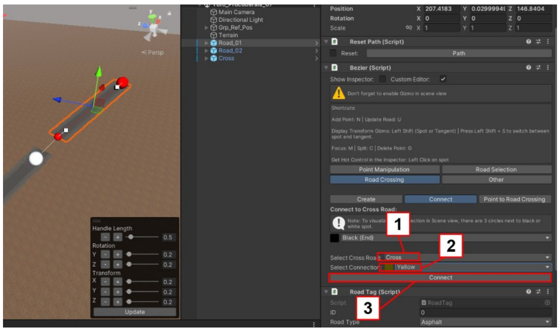
Press Connect button (spot 3)
Choose Black(End) button (spot 4)

Press Cross Road List button (spot 5)

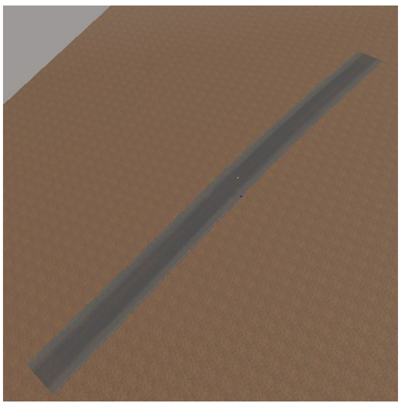


-In Inspector tab:
Choose Cross (spot 1)
Choose Yellow (spot 2)

Press Connect button (spot 3)



Road is now connected with crossroad



Create your own road texture and surface

This chapter explains how to create a road that is different from an asphalt road.

For that it is necessary to:

-Indicate the type of surface

-Change road and crossroad texture

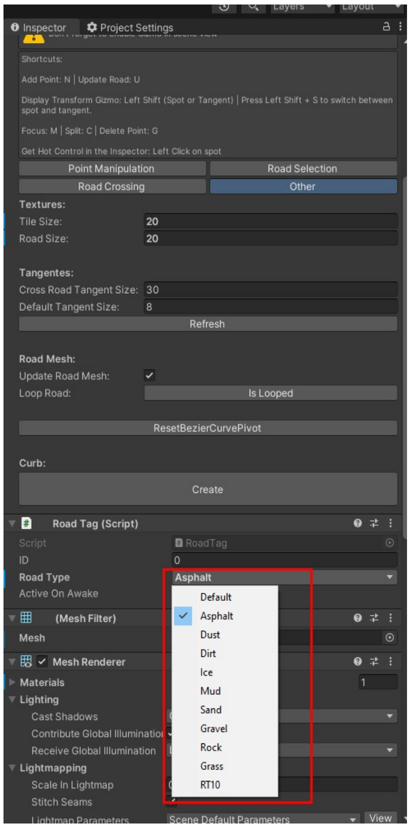
-Select a road

-In Inspector tab

Choose a surface from those proposed in the list.

<u>If your road is made up of several pieces and crossroads:</u>

Do the same thing on all parts.



-In project tab:
select Mat_Road_Asphalt

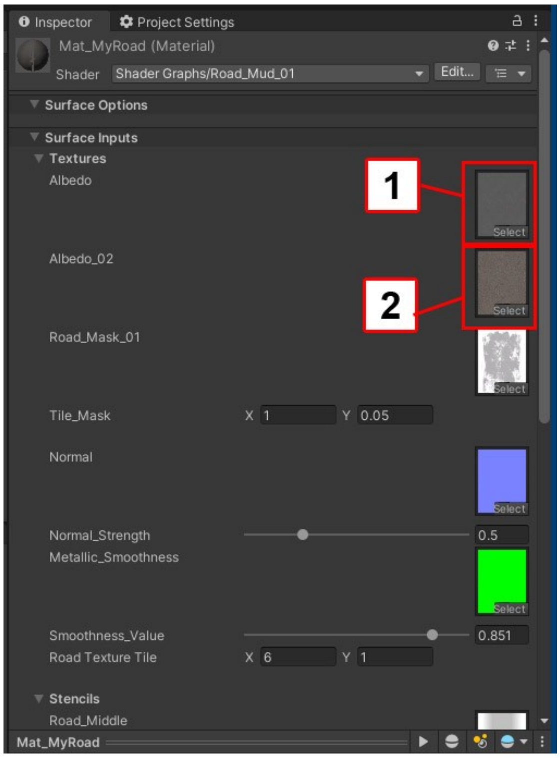
Assets > Environment > Materials> Roads

-In project tab:
Duplicate Mat_Road_Asphalt

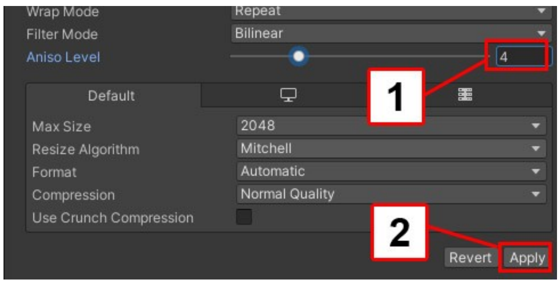
-Rename it for example Mat_MyRoad

-Drag and drop your textures in Albedo (spot 1) and Albedo_02 (spot 2)

<u>Info:</u>
Textures in Albedo and Albedo_02 can be the same or different



-Set the AnisoLevel of your texture to 4 (spot 1)
-Press button Apply (spot 2)



-In project tab:
select RoadCross
Assets > Environment > Materials> Roads

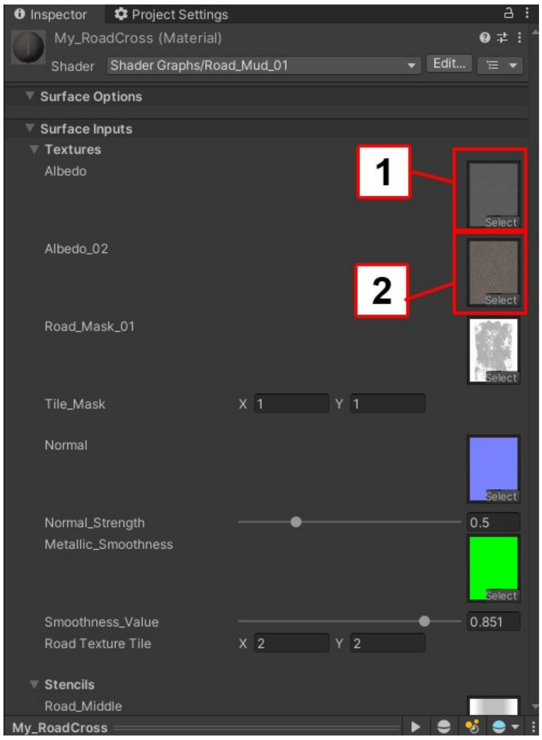
-In project tab:
Duplicate RoadCross

-Rename it for example My_RoadCross

-Drag and drop your textures in Albedo (spot 1) and Albedo_02 (spot 2)

Important :

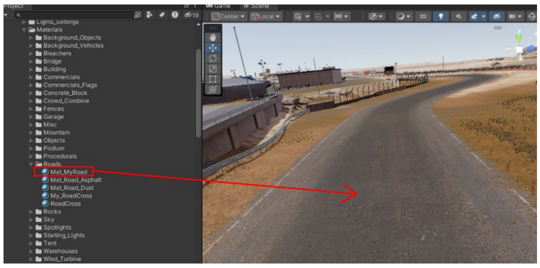
Use the same textures as used previously for Mat_MyRoad



-From Project tab:

drag and drop Mat_MyRoad onto your road (into the scene view)

Do the same with My_RoadCross if you have crossroads.

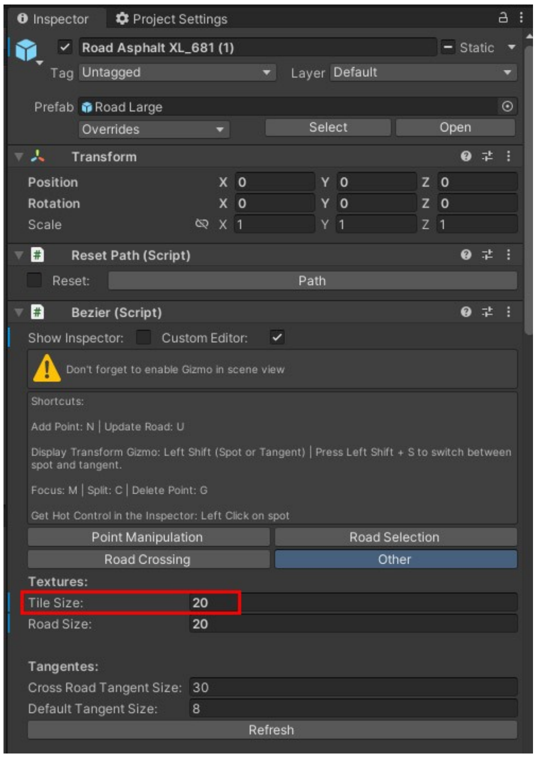


<u>If the road texture is too stretched:</u>

-Select the road

-In inspector tab press Other button.

-In Inspector tab set Tile Size to achieve the desired result



Procedurals Features

Overview

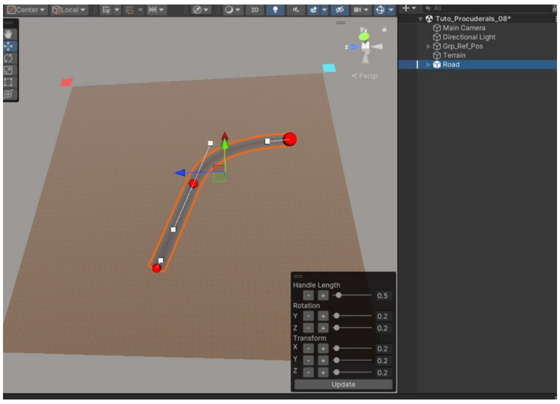
Important:
Procedural features use the same system as roads.
It is advisable to learn how to make the roads before learning how to create the procedural features.

Create Curb

In Project tab double click on Tuto_Procedurals_08 scene to open it.

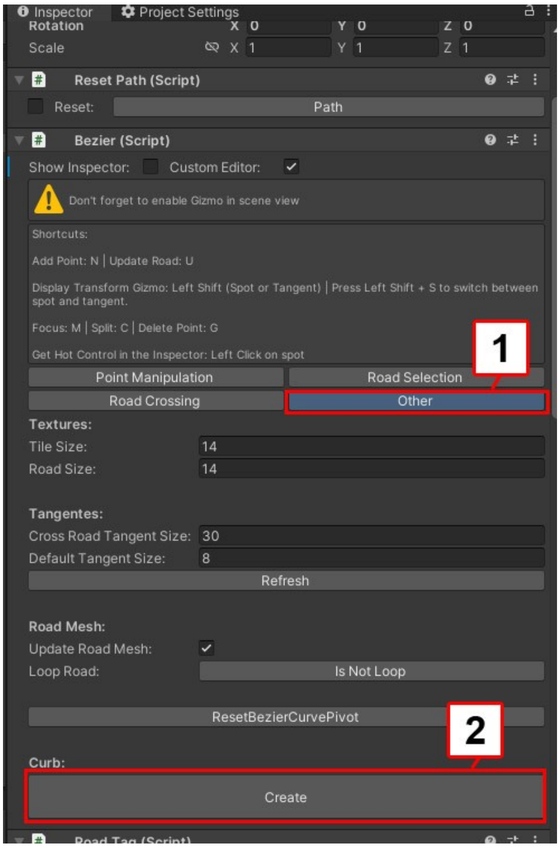
Assets > Scenes > Tutorials> Tuto_Procedurals

-In hierarchy tab select Road

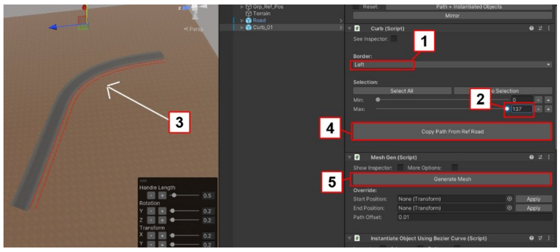


-In Inspector tab:
Press Other button (spot 1)

Press Create button (spot 2)



-In Inspector tab:
Choose Right or Left (spot 1)



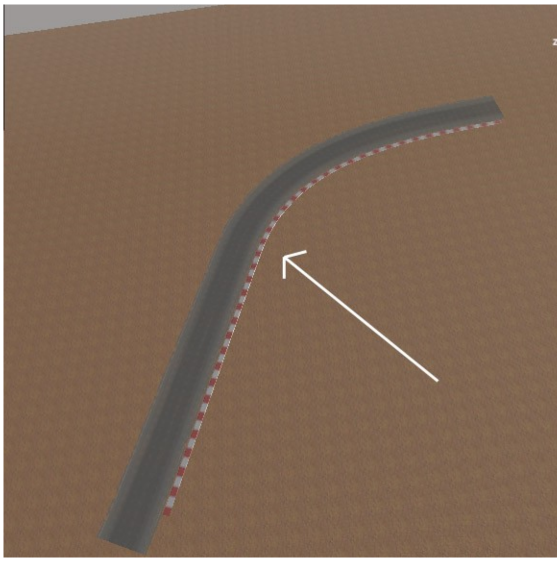
Set border length by changing start and end:
Set Max to 137 (spot 2)

In a scene view the red line indicates the size of the curbs (spot 3)

Press Copy Path From Ref Road button (spot 4)

Press Generate Mesh button (spot 5)

The curbs were created along the road



Create crash barrier

If you have not setup the procedural script read Link

In Project tab double click on Tuto_Procedurals_09 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

The curve system for creating crash barrier is the same curve system for creating roads.

For more informations about how to create roads Link

You can use the same keyboard shortcuts:

Split curve: select 2 points then press C

Delete Point: select point then press G

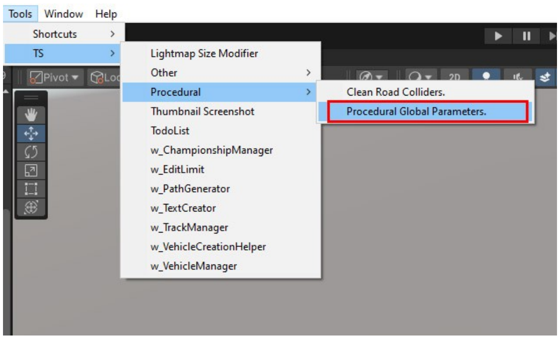
Active Handle solution 1: While holding down the Shift (Maj) press S then release S
While continuing to hold down the Shift (Maj), move the axis you want.

Active Handle solution 2: use the overlay tool
For more informations about the overlay tool Link

To select several points
-Select a Point
-While holding down the Left Ctrl select a second point

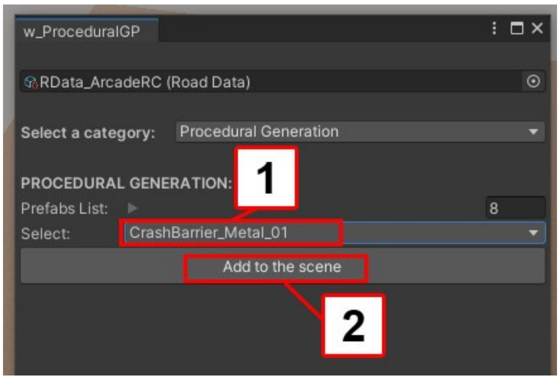
Important:
Don't forget to enable gizmos in scene mode if you don't.

Go to Tools > TS > Procedural > Procedural Global Parameters

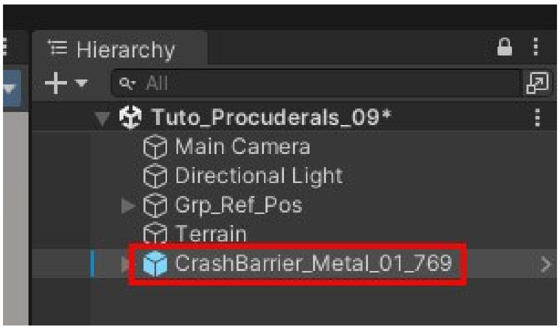


-In select list choose CrashBarrier_Metal_01 (spot 1)

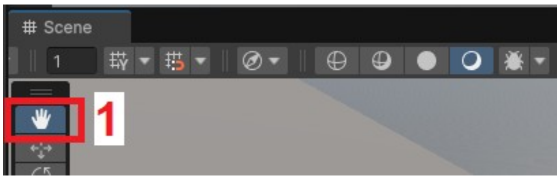
-Press Add to the scene button (spot 2)



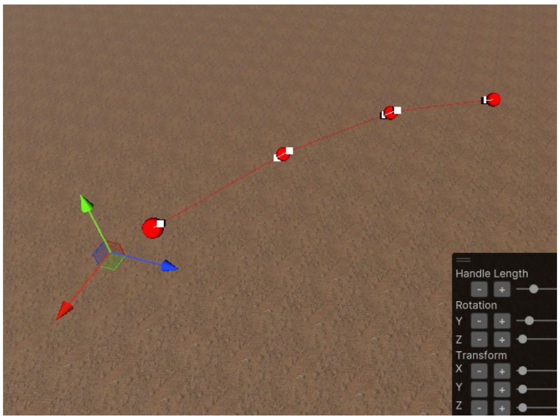
In hierarchy tab:
Select CrashBarrier_Metal



-Into scene view, to be sure there is the focused on Scene view:
Select View Tool (spot 1)
to force focus on scene view
Do this only the first time you create a point



In scene tab:
-Press keyboard shortcut N to create points

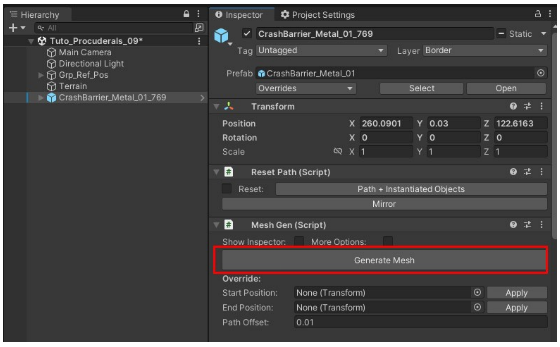


Important:
If the first point doesn't appear:

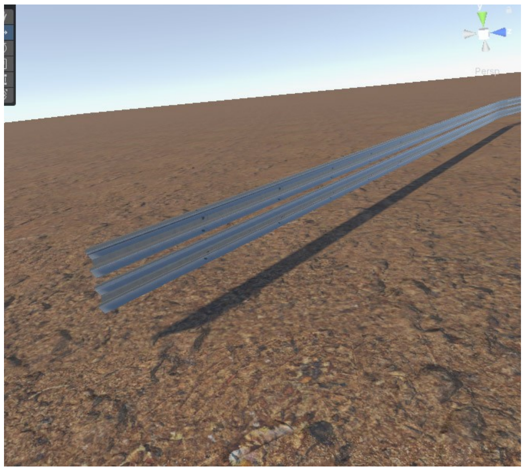
-Verify that gizmos are activate in scene tab
-In scene view select the terrain
-In hierarchy tab select again the new crash barrier
-Press keyboard shortcut N

In hierarchy tab:
Select CrashBarrier_Metal

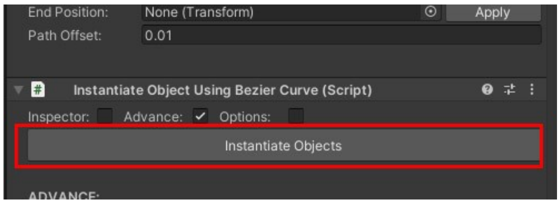
In Inspector tab:
Press Generate Mesh button



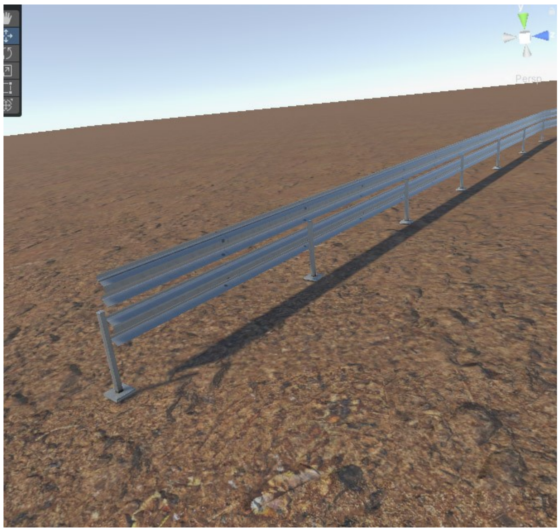
Barrier is created



In Inspector tab:
Press Instantiate Objects button

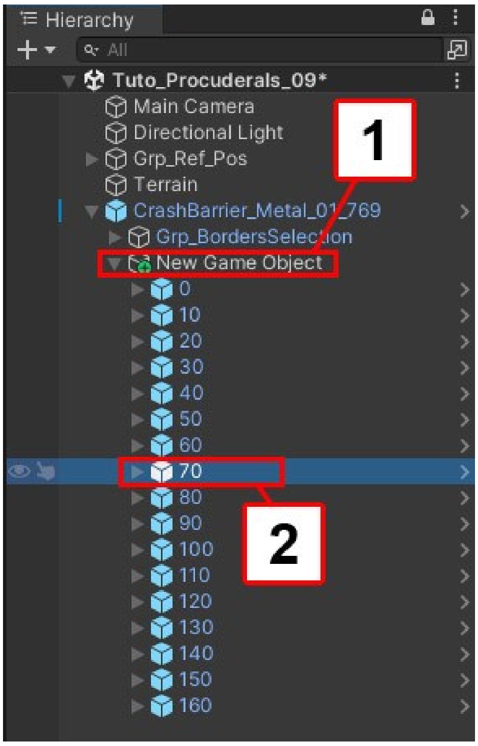


Plots are created



To delete a pole

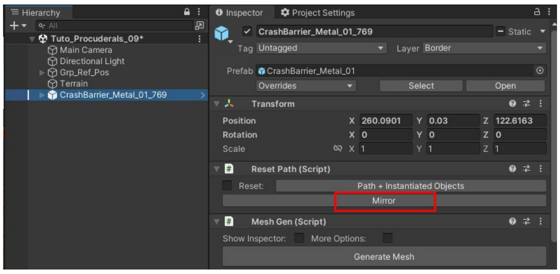
<u>In hierarchy tab:</u>
Into New Game Object group (spot 1):
-select a pole (spot 2)
-delete it



To reverse Crash barrier

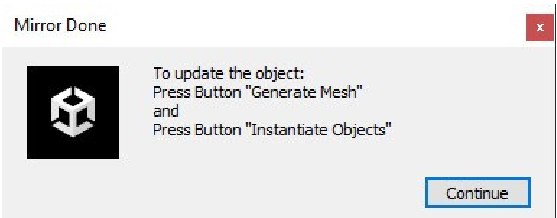
In hierarchy tab:
Select CrashBarrier_Metal

In Inspector tab press Mirror Button



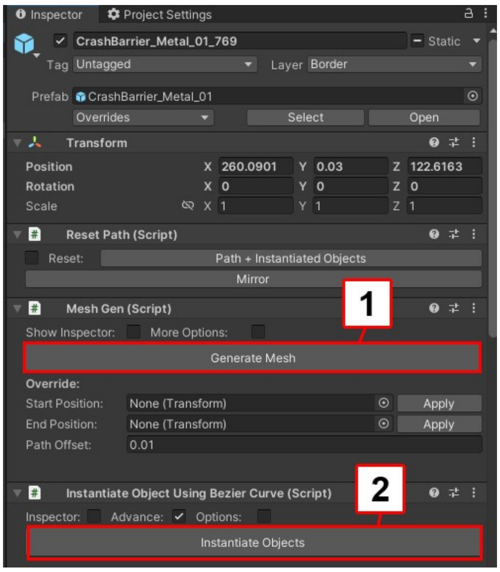
A new window appears

Press Continue button



In Inspector tab:
Press Generate Mesh button to mirror the barrier (spot 1)

Press Instantiate Objects button to mirror the plots (spot 2)



Create wall + metal fence

Wall + metal fence works the same as Crash barrier

For more informations about how to create Crash barrier Link

Create Barrier Wood

If you have not setup the procedural script read Link

In Project tab double click on Tuto_Procedurals_09 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

The curve system for creating Barrier Wood is the same curve system for creating roads

For more informations about how to create roads Link

You can use the same keyboard shortcuts:

Split curve: select 2 points then press C

Delete Point: select point then press G

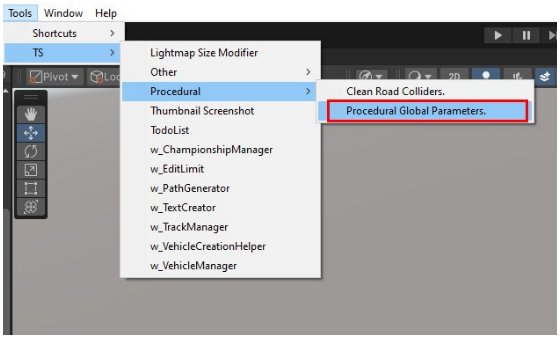
Active Handle solution 1: While holding down the Shift (Maj) press S then release S
While continuing to hold down the Shift (Maj), move the axis you want.

Active Handle solution 2: use the overlay tool
For more informations about the overlay tool Link

To select several points
-Select a Point
-While holding down the Left Ctrl select a second point

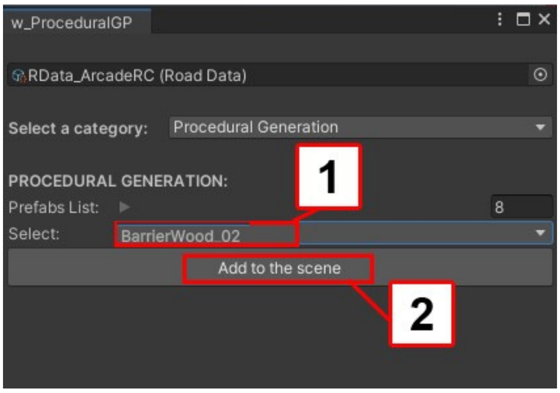
Important:
Don't forget to enable gizmos in scene mode if you don't.

Go to Tools > TS > Procedural > Procedural Global Parameters

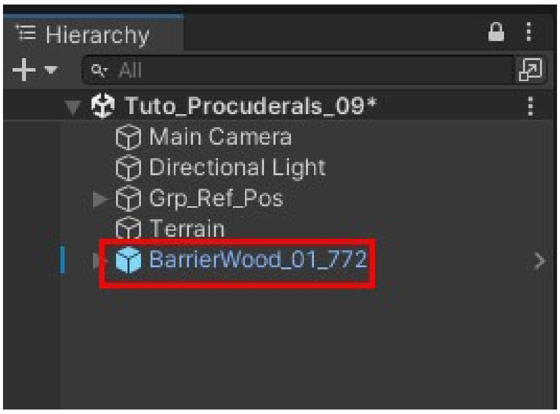


-In select list choose BarrierWood_02 (spot 1)

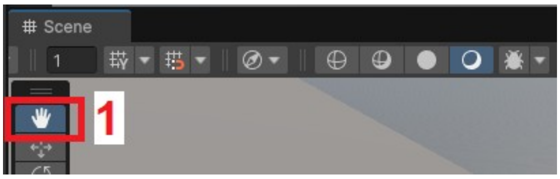
-Press Add to the scene button (spot 2)



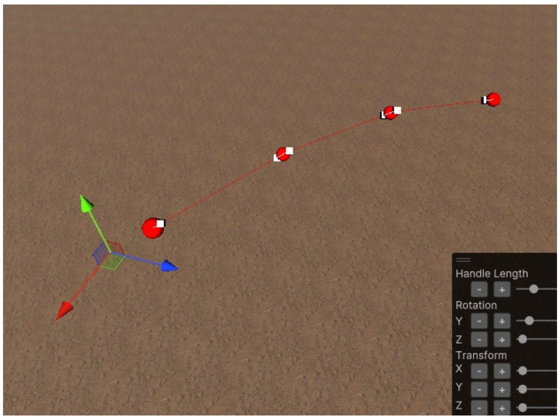
In hierarchy tab :
Select BarrierWood



-Into scene view, to be sure there is the focused on Scene view:
Select View Tool (spot 1)
to force focus on scene view
Do this only the first time you create a point



In scene tab:
-Press keyboard shortcut N to create points

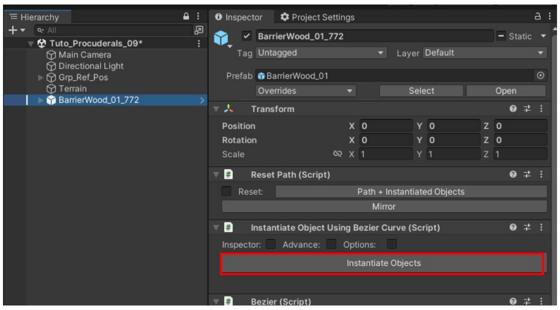


Important:
If the first point doesn't appear:

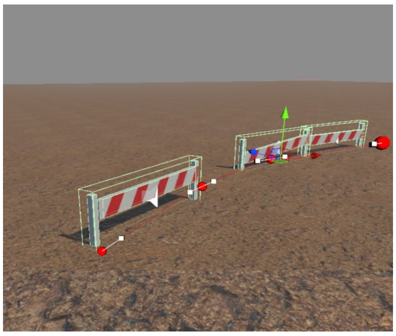
-Verify that gizmos are activate in scene tab
-In scene view select the terrain
-In hierarchy tab select again the new wood barrier
-Press keyboard shortcut N

In hierarchy tab:
Select BarrierWood

In Inspector tab :
Press Instantiate Objects button



Barrier are created



Create roads blocks

If you have not setup the procedural script read Link

In Project tab double click on Tuto_Procedurals_09 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

The curve system for creating roads blocks is the same curve system for creating roads

For more informations about how to create roads Link

You can use the same keyboard shortcuts:

Split curve: select 2 points then press C

Delete Point: select point then press G

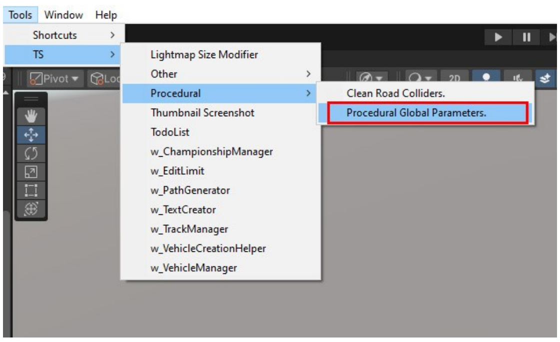
Active Handle solution 1: While holding down the Shift (Maj) press S then release S
While continuing to hold down the Shift (Maj), move the axis you want.

Active Handle solution 2: use the overlay tool
For more informations about the overlay tool Link

To select several points
-Select a Point
-While holding down the Left Ctrl select a second point

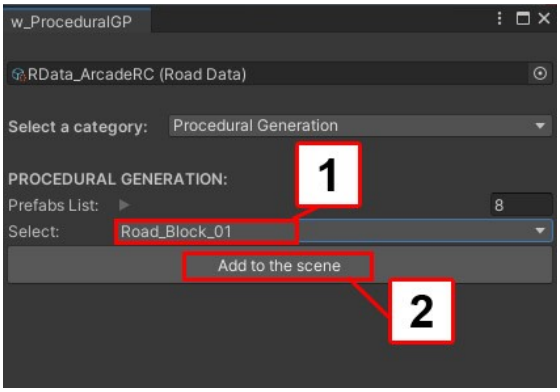
Important:
Don't forget to enable gizmos in scene mode if you don't.

Go to Tools > TS > Procedural > Procedural Global Parameters

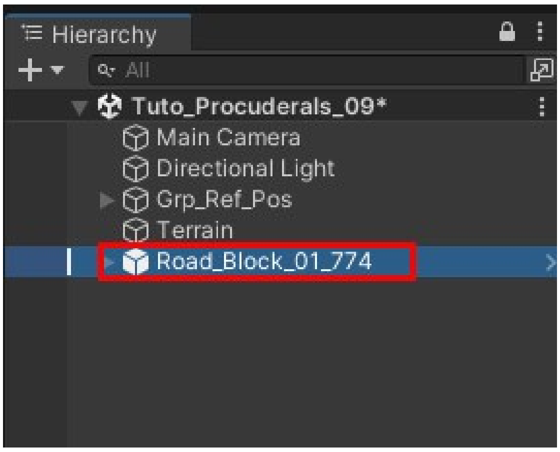


-In select list choose Road_Block_01 (spot 1)

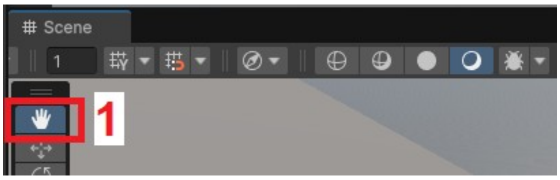
-Press Add to the scene button (spot 2)



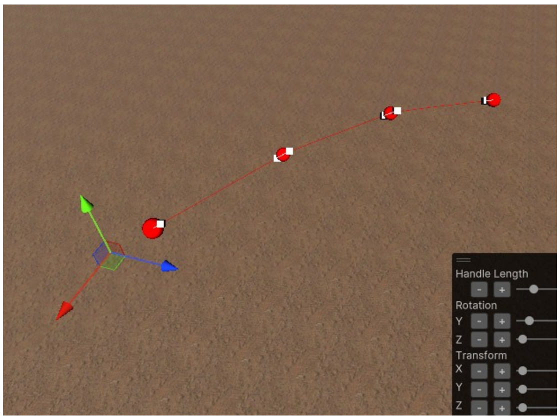
In hierarchy tab:
Select Road_Block



-Into scene view, to be sure there is the focused on Scene view:
Select View Tool (spot 1)
to force focus on scene view
Do this only the first time you create a point



In scene tab:
-Press keyboard shortcut N to create points

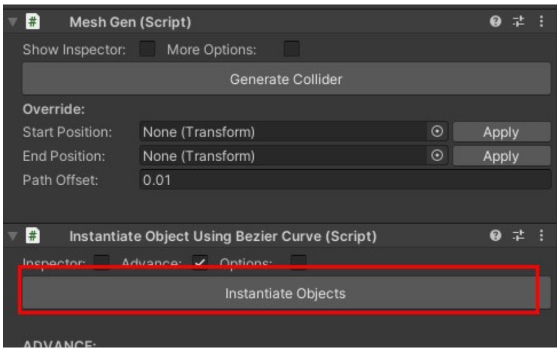


Important:
If the first point doesn't appear:

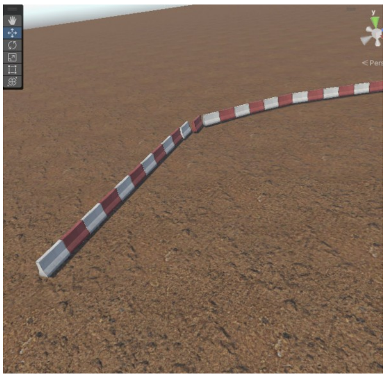
-Verify that gizmos are activate in scene tab
-In scene view select the terrain
-In hierarchy tab select again the new road bock
-Press keyboard shortcut N

In hierarchy tab:
Select Road_Block

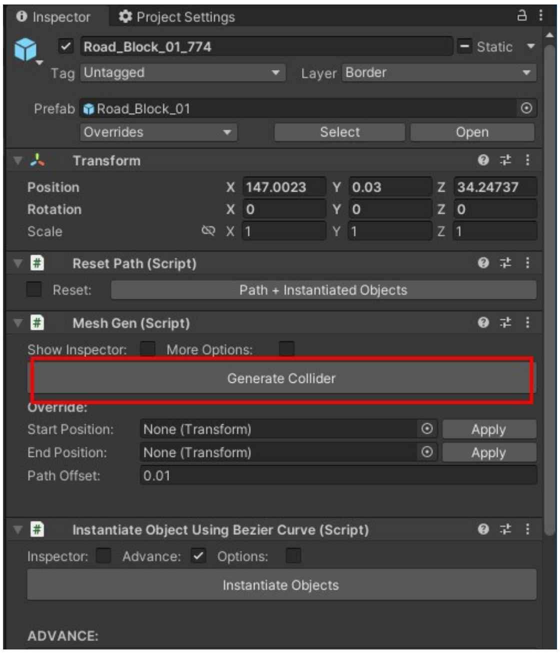
In Inspector tab:
Press Instantiate Objects button



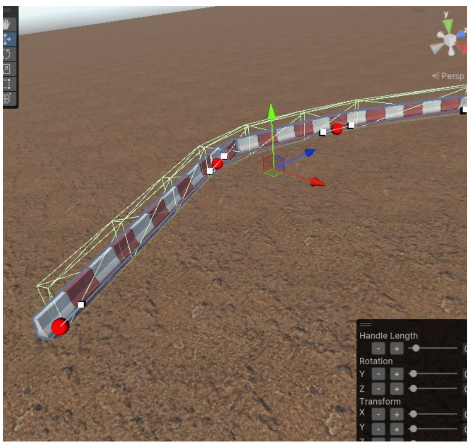
Road_Block are created



In Inspector tab:
Press Generate Collider button



Collider are created



Create special collider

If you have not setup the procedural script read Link

In Project tab double click on Tuto_Procedurals_09 scene to open it.

Assets > Scenes > Tutorials> Tuto_Procedurals

The curve system for creating Special Collider is the same curve system for creating roads

For more informations about how to create roads Link

You can use the same keyboard shortcuts:

Split curve: select 2 points then press C

Delete Point: select point then press G

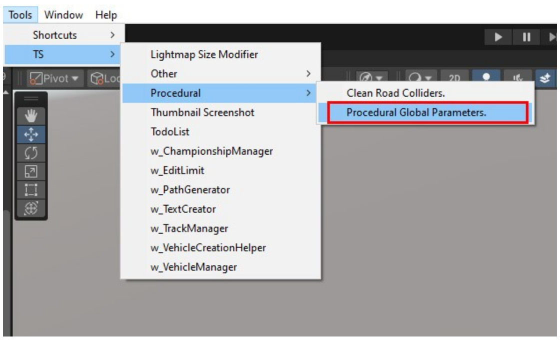
Active Handle solution 1: While holding down the Shift (Maj) press S then release S
While continuing to hold down the Shift (Maj), move the axis you want.

Active Handle solution 2: use the overlay tool
For more informations about the overlay tool Link

To select several points
-Select a Point
-While holding down the Left Ctrl select a second point

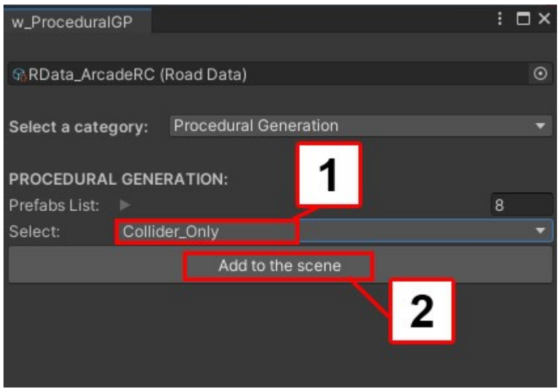
Important:
Don't forget to enable gizmos in scene mode if you don't.

Go to Tools > TS > Procedural > Procedural Global Parameters



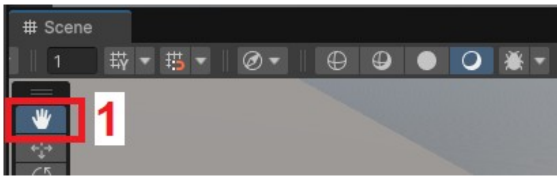
-In select list choose Collider_Only (spot 1)

-Press Add to the scene button (spot 2)



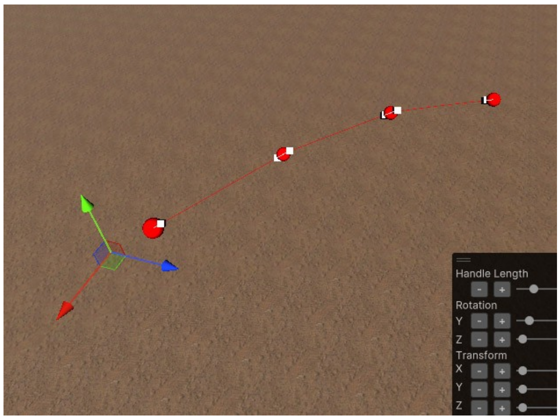
In hierarchy tab:
Select Collider_Only

-Into scene view, to be sure there is the focused on Scene view:
Select View Tool (spot 1)
to force focus on scene view
Do this only the first time you create a point



-Into scene view:
Mouse left click to activate scene view
Do this only the first time you create a point

In scene tab:
-Press keyboard shortcut N to create points

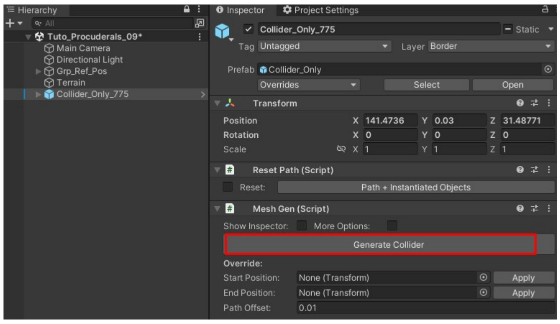


Important:
If the first point doesn't appear:

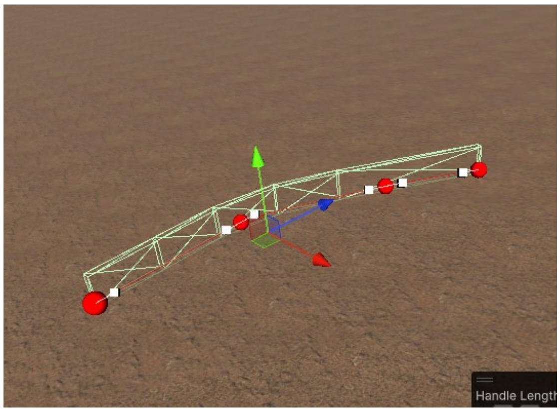
-Verify that gizmos are activate in scene tab
-In scene view select the terrain
-In hierarchy tab select again the new Collider
-Press keyboard shortcut N

In hierarchy tab:
Select Collider_Only

In Inspector tab :
Press Generate Collider button



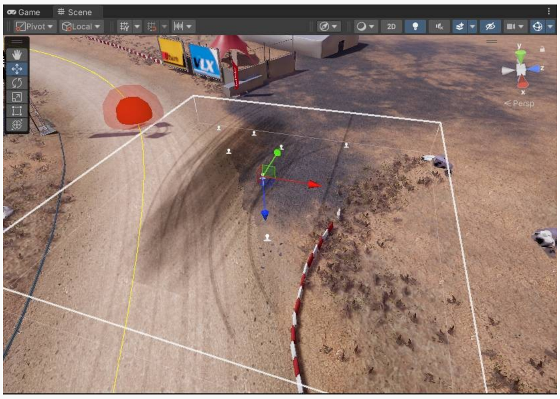
Collider is created



Decal

How it works

Unity Definition: With the Decal Renderer Feature, Unity can project specific “texture” onto other objects in the Scene. The decals interact with the Scene’s lighting and wrap around Meshes.



By default, the Decal applies to objects within its area. This area is represented by the white cube that appears when selecting a decal.

This is a problem in many cases.

For example, for tire tracks on the ground:

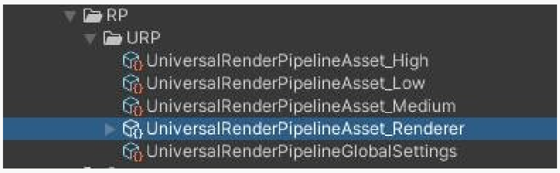
We want the decal to apply to the ground, but we don't want the decal to apply to the car when it enters the decal's area.

To solve this problem Unity offers a system to apply the decal only to certain objects using a layer system.

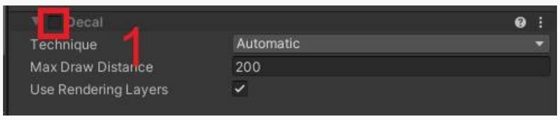
How to: Disable Decal Feature

By default the Decal feature is already enabled.
<u>If you don’t want to use decals:</u>

-In Project tab select UniversalRenderPipelineAsset_Renderer



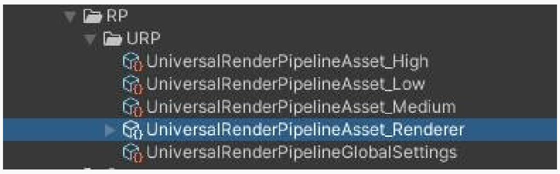
- In the Inspector uncheck Decal Feature toggle (spot 1).



How to: Add Decal Feature

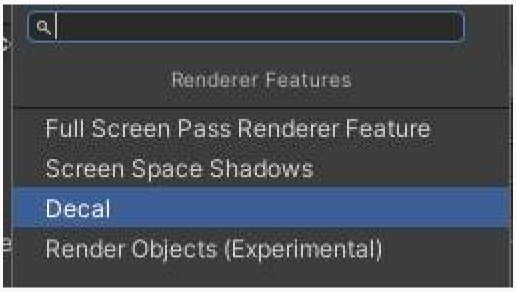
For information Decal feature is already setup in the asset. But for any reason if you want to setup decals read the following tutorials

- In Project tab select UniversalRenderPipelineAsset_Renderer

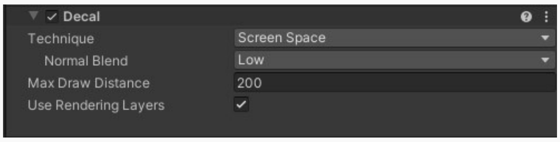


- In the inspector press Add Feature Renderer button

- Select Decal in the list.



Decal is added to the Renderer Features list.



Technique: ScreenSpace mode must be selected if the terrain uses more than 4 terrain layers.

Max Draw distance corresponds to distance to display the Decal. The smaller the value, the more you limit the impact on game performance.

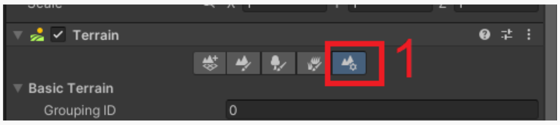
Check Use Rendering Layers to be able use the layer system and apply decals only to certain objects in the scene.

How to: Setup to be available to receive Decal

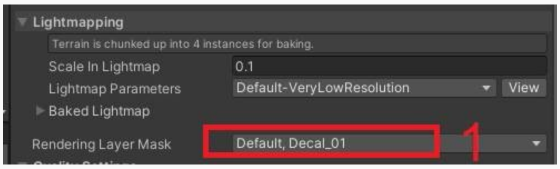
Case 1: Unity Terrain

- Select the terrain you want to apply Decals to.

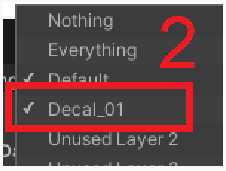
- In the Inspector go to Terrain Setting tab (spot 1).



- Go to the bottom of the Inspector and open the dropdown menu named Rendering Layer Mask (spot 1).



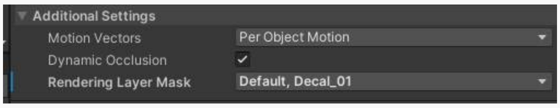
- Select Layer Decal_01 (Spot 2).



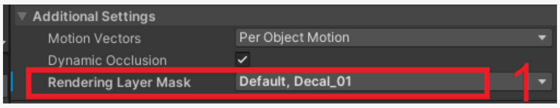
Case 2: Object

- Select the object you want to apply Decals to.

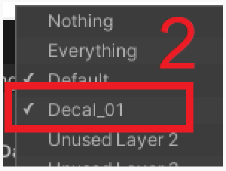
- In the Inspector go Additional Settings.
Open it if needed.



- In Inspector open the dropdown menu named Rendering Layer Mask
(spot 1).



- Select Layer Decal_01 (Spot 2).



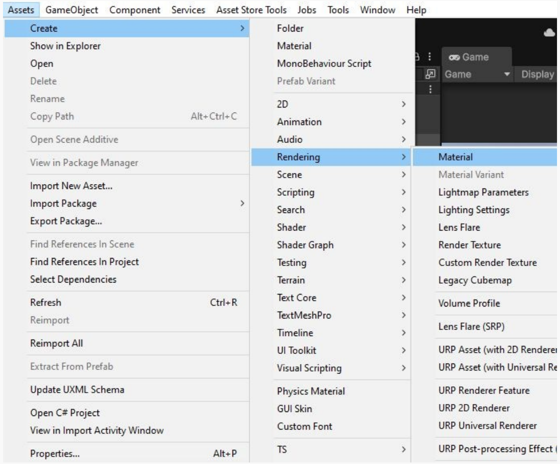
How to: Create a Decal

Ready-to-use decals are included in folder Decals

Assets → Environment → Decals→ Prefabs

If you want to add your own decals you can follow the tutorial below:

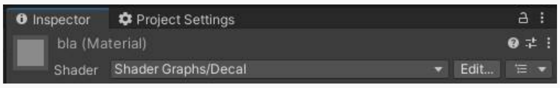
- In the Project tab create a material:
Menu → Assets → Create → Rendering → Material



A new decal is created in your Project Tab.

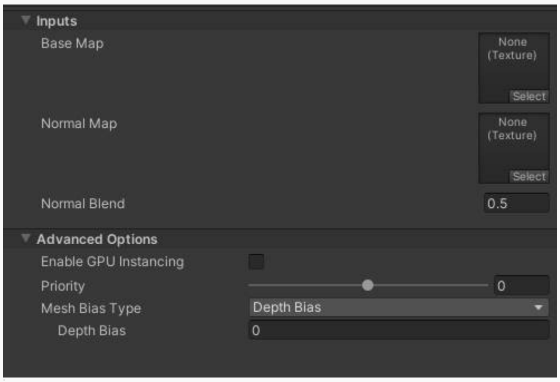
- Rename it. For this example rename it MyDecalMat

- In the Inspector select the shader
Shader Graph/ Decal



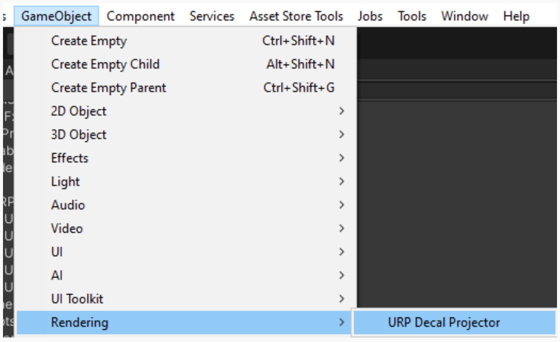
- Add your texture in the base map.

- Add your normal map if needed.

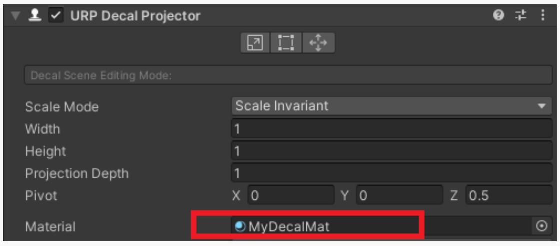


- Create a URP Decal Projector.
Menu → GameObject → Rendering → URP Decal projector

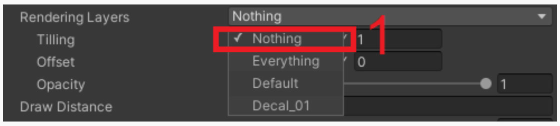
A new decal is created in your Hierarchy.



- In the Inspector change the default material with the new one we’ve created.



- In the Rendering Layer choose first choose Nothing (spot 1) Decal_01



- Then choose first choose Decal_01

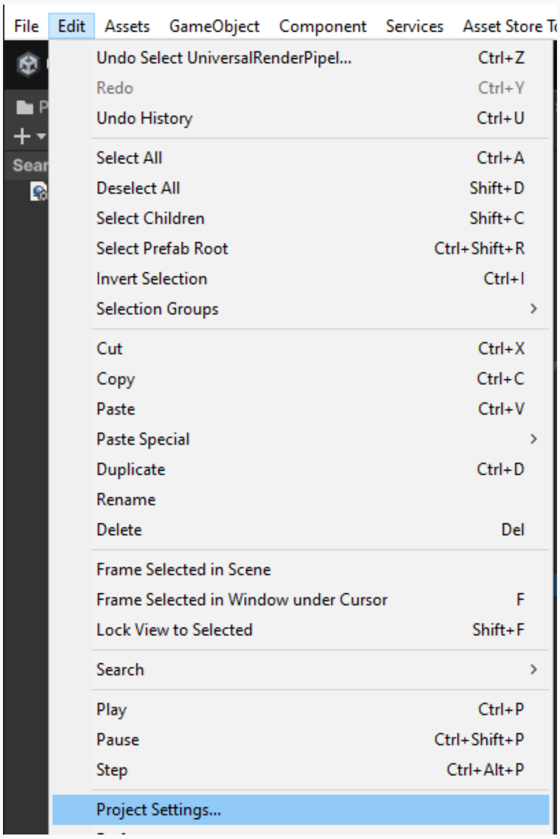
You should have this result.

Now you can move the decal in the scene. The decal will be projected on the objects having selected the Decal_01 layer in Rendering Layer Mask.

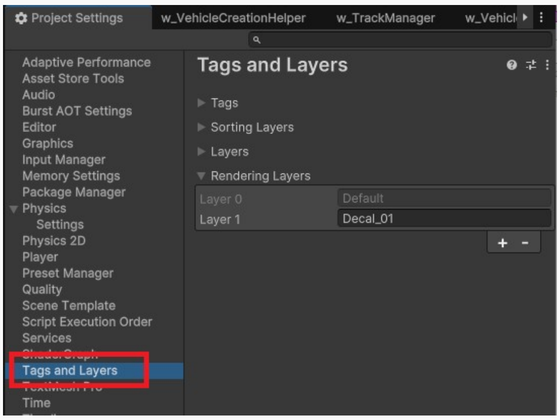
How to: Create a layer for Decal

Note:
Note: The more layers you add, the more you impact game performance. Try to limit the number of layers

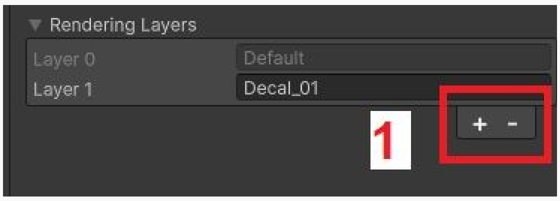
- Open the Project Settings:
Edit → Project Settings



- On the left panel click on Tag and Layers.

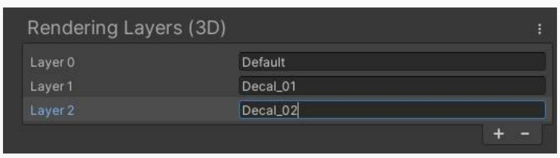


- In the Inspector, in section Rendering Layers press + button (spot 1) to create a new Layer.



- Rename the layer. For example Decal_02.

Now you can use this layer.

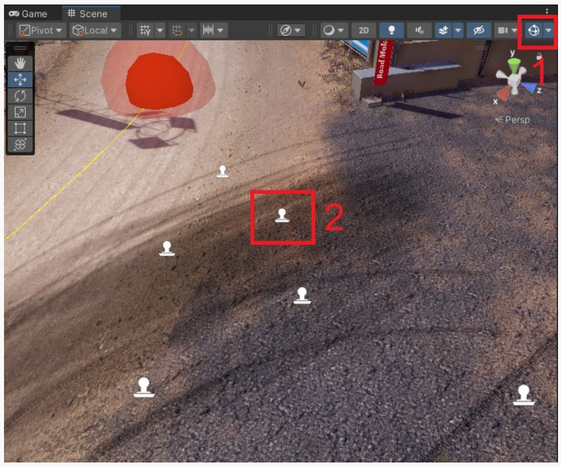


How to: Manipulate Decal

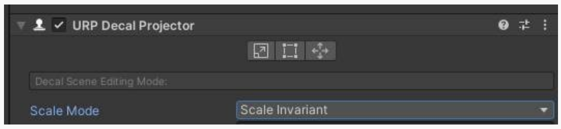
This section gives you some tips to manipulate Decals.

To be able to select a Decal in the scene view:
- In the Scene View enable the Gizmos by pressing the gizmos button (spot 1) *(Top right of window)*

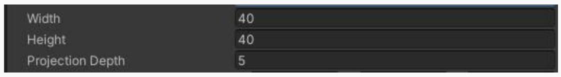
-In the scene press a Decal icon (spot 2) to select a decal. When the decal is selected you can move it in the scene.



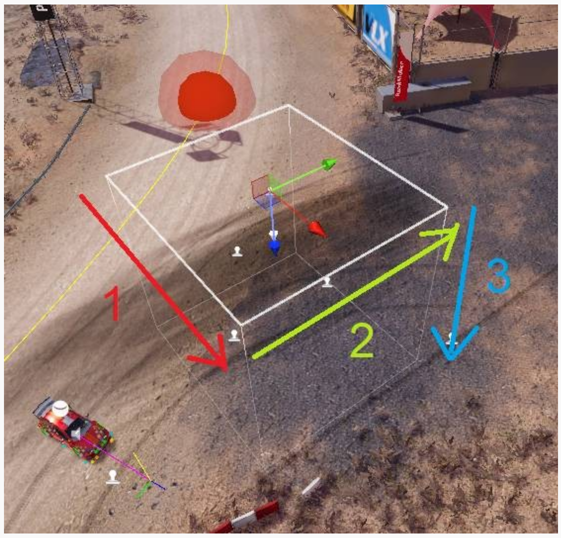
- In the Inspector you can modify some parameters:



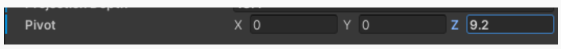
The first 3 parameters allows to choose the size of the area where the Decal is applied to objects.



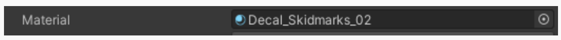
Width (spot 1)
Height (spot 2)
Projection Height (spot 3)



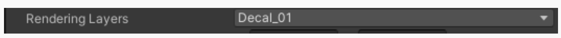
Change the Decal pivot.



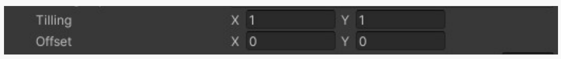
The material that contained the Decal texture.



Object with this layer is affected by this Decal.

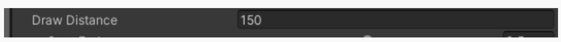


You can change the Decal Tile and offset.



Choose the Decal opacity.

Choose the max distance to see the Decal.

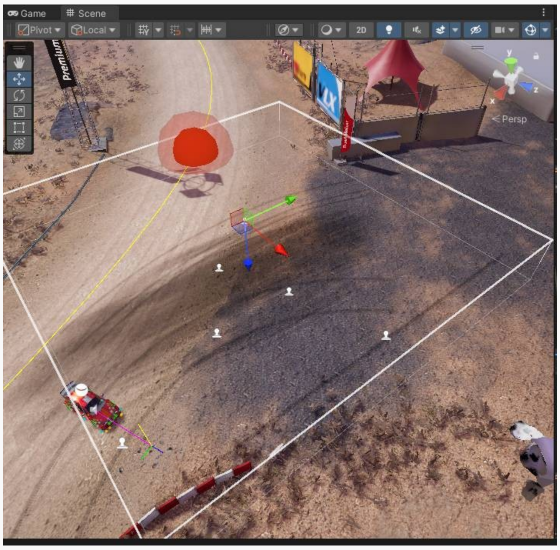


Fade the Decal depending the distance to the camera.

Reminder:

A decal is projected on an object when:
- The object and the Decal are using the same Rendering Layer Mask.

A decal is projected on an object when:
- The object is inside the Decal area. The decal area is represented by a white cube.



The projection will be better if the object and the decal are parallel. The less parallel they are, the more the decal looked distorted.

Ready to Use Decals

Ready-to-use decals are included in folder Decals > Prefabs

Assets → Environment → Decals→ Prefabs

Lighting And Rendering

Reflection Probes

A reflection probe must encompass the entire scene

Global reflection prob is included in the starter kit scene

Light Probes

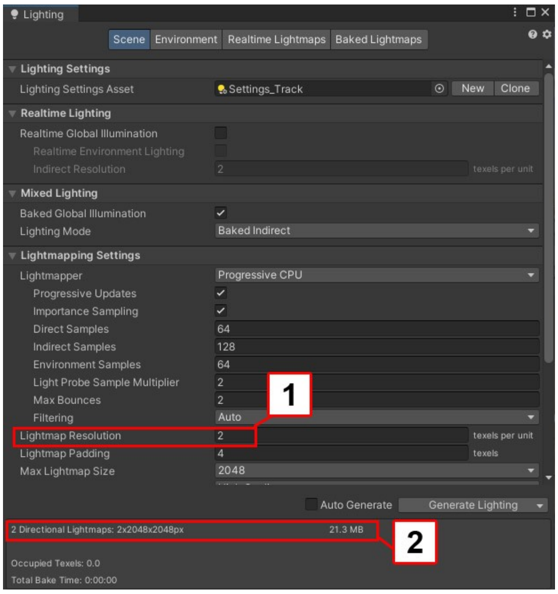
A Lightprobe Group must be add to the scene.
Lightprobe are used to light moving objects.

Lights Probes are included in the starter kit scene

Lightmaps

Lightings tab presets are ready to use in the starter kit

Info about Lightmaps parameters:



Demo scene is large then a low Lightmap Resolution is used (spot 1)

Increase the Lightmap resolution value increase the quality of the lightmaps but:
-Lightmaps computation is longer
-Lightmaps size are bigger

The size of the lightmaps are visible at the bottom of the lighting tab (spot 2)

In Lighting tab Lightmap Resolution parameter is set to 2.

A value of 2 is good for most of elements.
But some objects may needs more definition.

It is therefore necessary to increase the definition of lightmaps, but only for this objects.

To increase the quality of lightmaps of a particular object, you must increase its scaleInLighmap parameter.
Changing the scaleInLightmap value of each object would be very tedious in particular with LODs objects
This is why a script **is included in the asset**, to modify the scaleInLightmap parameter of a set of objects with one click.

To learn how to use the script: [Link]

Lightmap calculation time

The calculation time of the track scene is about 10 minutes
Size of Lightmaps are about 35 MB

The calculation of lightmaps has been tested with this configuration:

PC configuration for test:
11th Gen Intel(R) Core(TM) i7-11700
RAM 16,0 Go
Nvidia Geforce RTX 3060 Ti

Info:
Do not rely on the time indicated at the beginning of the calculation of the Lightmaps. The calculation time is much shorter than the times indicated at the beginning of the calculation.

Post Effects

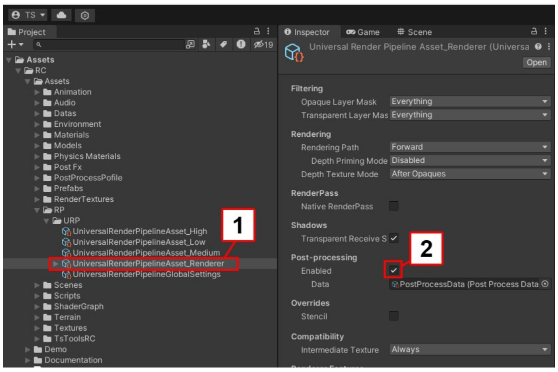
Pipeline Setup

PostFx **must** be enabled. If you haven't done it yet:
In Project settings tab:

-Select UniversalRenderPipelineAsset_Renderer (spot 1)

Assets > RP > URP

-Check Enabled Checkbox (spot 2)



Global PostFx

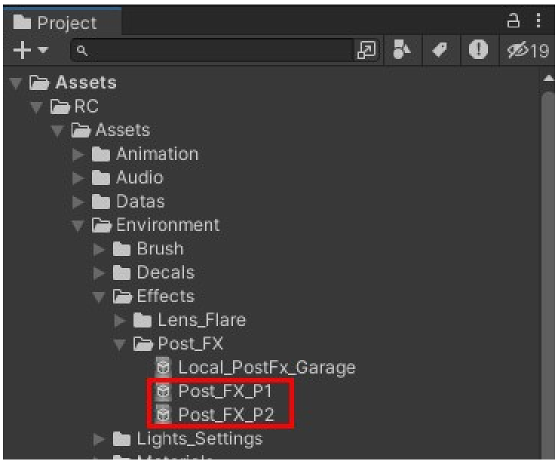
PostFx are in Post_FX folder

Assets > Environment > Effects > Post_FX

Post_FX_P1 are used by player 1 camera

Post_FX_P2 are used by player 2 camera

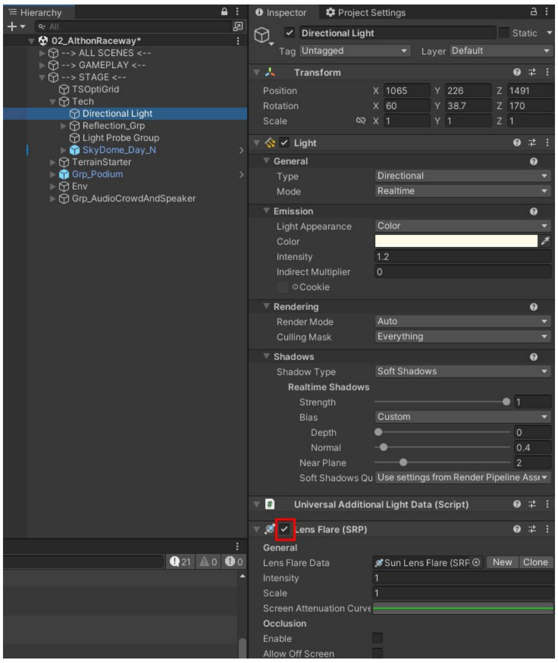
<u>If you modify Post_FX_P1:</u>
don't forget to do the same modification to Post_FX_P2



Lens Flare

There is a lens flare component on Directional Light (Sun)

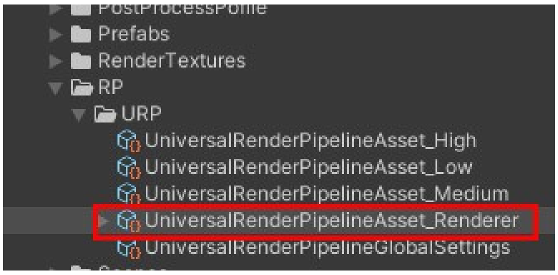
<u>If you don’t want to use it:</u>
-Select Sun directional light:
-In Inspector tab uncheck Lens Flare (SRP) checkbox.



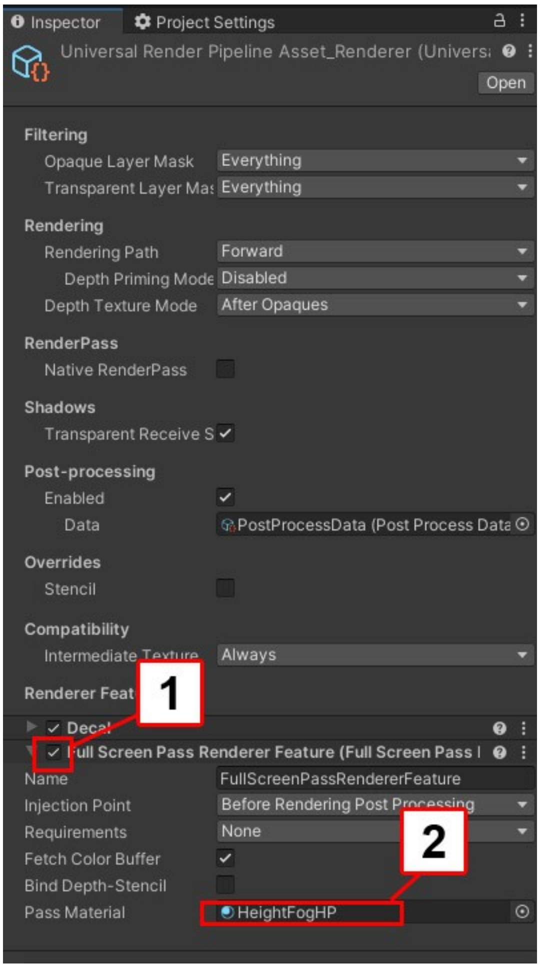
Depth Fog (Custom PostFx)

The script allows you to modify the color, contrast and saturation depending on the distance (Zdepth).
It's a custom postFx

In project tab select
UniversalRenderPipelineAsset_Renderer



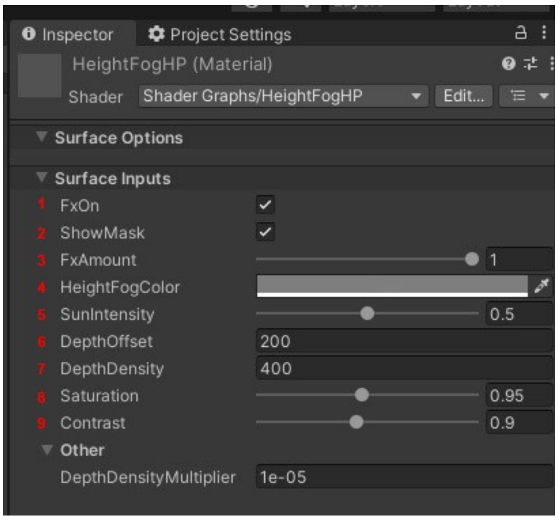
In Inspector:
- check Full screen Pass RenderFeatures checkbox to activate the custom postFX (spot 1)
-press HeightFogHP to select it in project tab (spot 2)



In project tab select HeightFogHP

Parameters:

1 FxOn :Activate the custom postFx
2 ShowMask: Show the Zdepth mask
3 FxAmount: Intensity of the postFx
4 HeightFogColor: Add Color depending on distance
5 SunIntensity: Intensity of sunlight
6 DepthOffset: Zdepth mask distance
7 DepthDensity: Zdepth mask density
8 Saturation: Saturation depending on distance
9 Contraste: Contraste depending on distance



Optimization

Overview

Large environment needs optimization.
This is why optimization scripts are included in the asset.

Optimization system

Overview:
A script is included in the asset that makes some of the objects hide or unhide depending on the distance from the player position.

For whatever reason if you don't want to use optimization script read: Link

The script is separated in 2 cases.
Depending on the position and size of objects, case 1 or case 2 will be chosen.
The two cases are used together.

Case 1: grid system
Objects appear or disappear depending on the car player position on the grid.
This system is ideal for small to medium sized objects.

For large objects use case 2 instead.

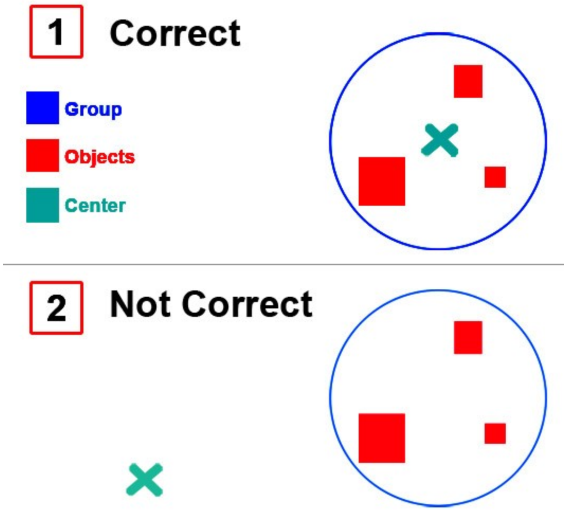
Case 2: group distance system
A distance is assigned to a group (for example 500 meters)
Objects included in this group will be visible within 500 meters.
This case is ideal for large objects.
Each group can have a different distance setup.

Small objects must be visible at 50 or 100 meters.
Medium or large objects must be visible at 300 or 400 meters.

Very Important:

To determine the distance, the optimization system takes into account the center of the group

In image 1:
The center of the group is in the middle of the objects.
This position is correct



In image 2:
The center of the group is not in the middle of the objects.
This position is incorrect

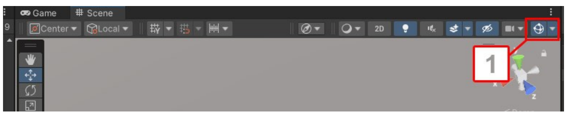
If the center of the group is not correctly placed it is possible that certain objects:
-do not appear
-appear/disappear at the wrong time

Case 1: Grid

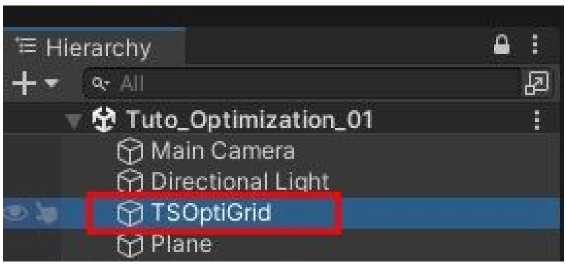
1 In Project tab double click on Tuto_Optimization_01 scene to open it

Assets > Scenes > Tutorials > Tuto_Optimization

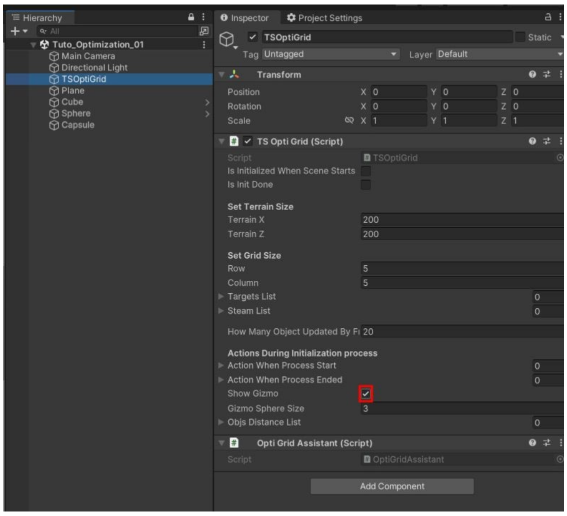
2 Click on Gizmo icon to activate gizmos visibility
(spot 1)



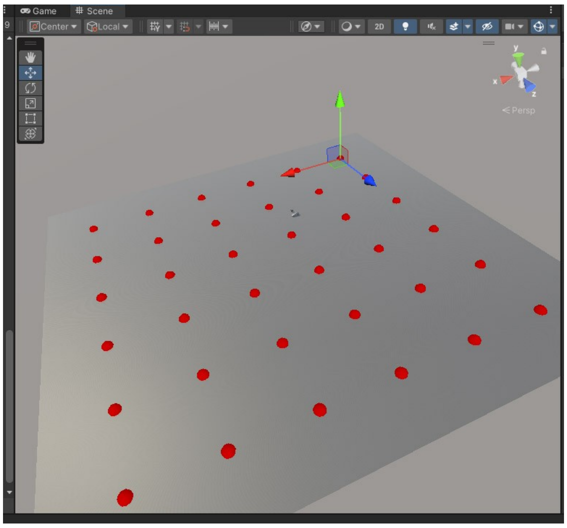
3 In hierarchy tab select TSOptiGrid



In Inspector tab check Show Gizmo checkbox

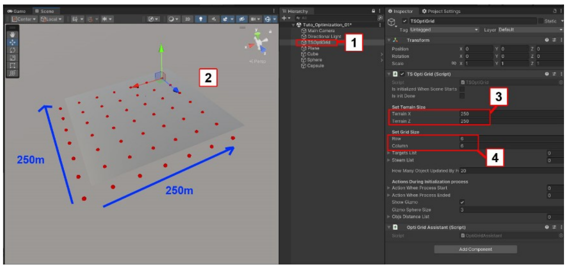


Grid dots are visible when TSOptiGrid
 is selected



In hierarchy tab:
-select TSOptiGrid (spot 1)

<u>Depending on needs it is possible to modify:</u>
-the size of the grid
-the number of dots on the grid (spot 2)



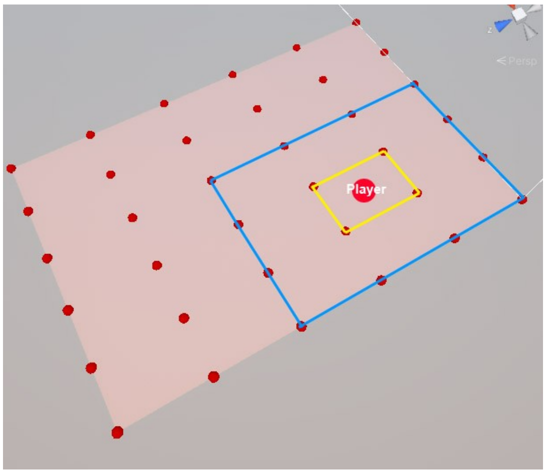
In Inspector tab:
-set Terrain X to 250
-set Terrain Y to 250 (spot 3)

-set Row to 6
-set Column to 6 (spot 4)

How it works

The car player is in yellow area.
Blue aera consists of the 9 zones around the yellow zone.
Objects that are in blue area are activated
Objects that are not in this area are deactivated

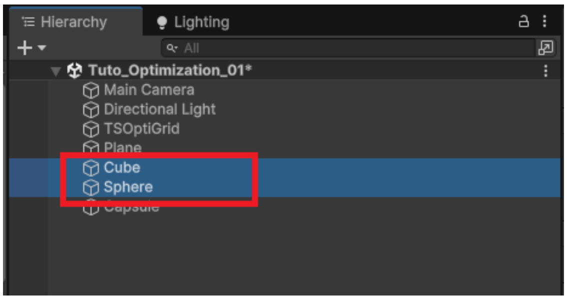
Set row and column according to your needs



Important:
For the script to work, the objects must be in a group that has the script TSStreamGridTag

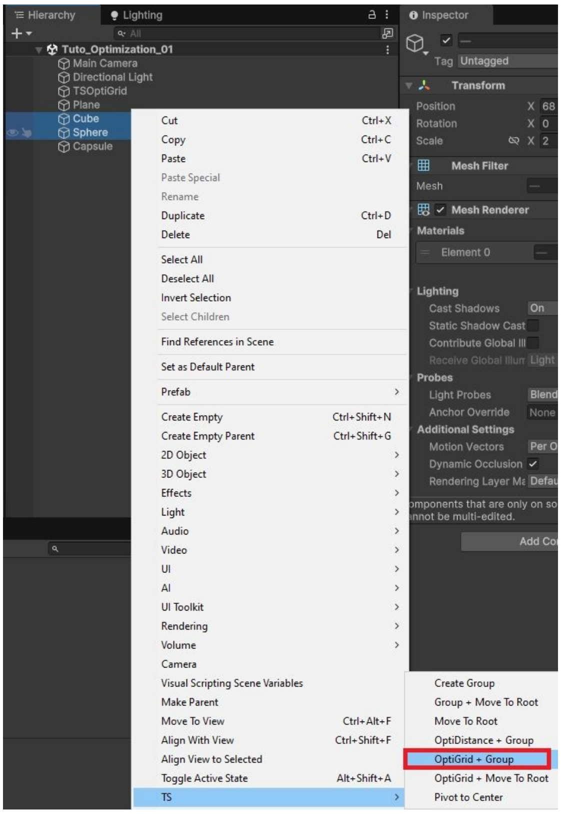
<u>Add TSStreamGridTag script to group:</u>

In hierarchy tab:
-select:
Cube
Sphere



-Mouse right click
-In the menu choose TS > OptiGrid + Group

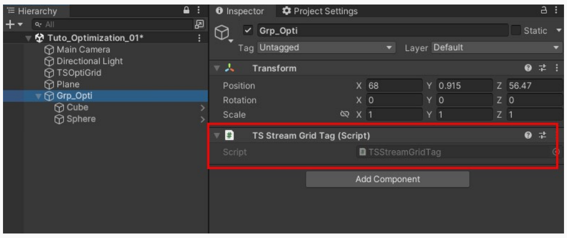
A new group is created.
Script TSStreamGridTag is added to the group



In hierarchy tab:
-select Grp_Opti

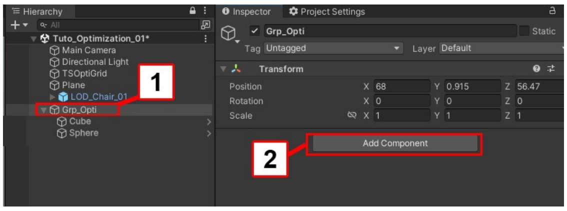
Script TSStreamGridTag is added to the group

Now this group works with the grid optimization system



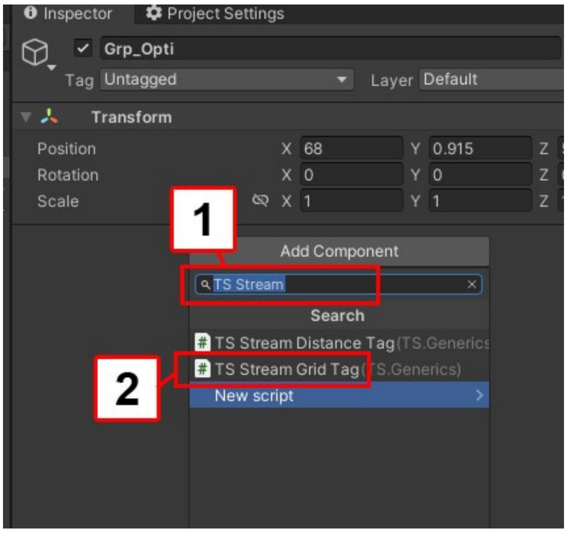
There is another way to add the script to a group:

-In hierarchy tab select a group (spot 1)
-Press Add component button (spot 2)



-In search field type TS Stream (spot 1)

-Choose TS Stream Grid Tag (spot 2)



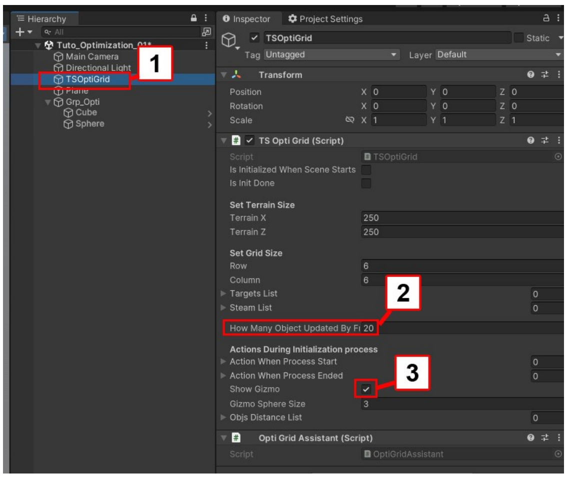
Optimization grid fine tuning

In hierarchy tab:
-select TSOptiGrid (spot 1)

Important:
OptiGrid group activated or deactivated 20 objects by frame.This avoids framerate drops.

-If you want to activated more or less objects by frame set How Many Objects Update By Frame value (spot 2)

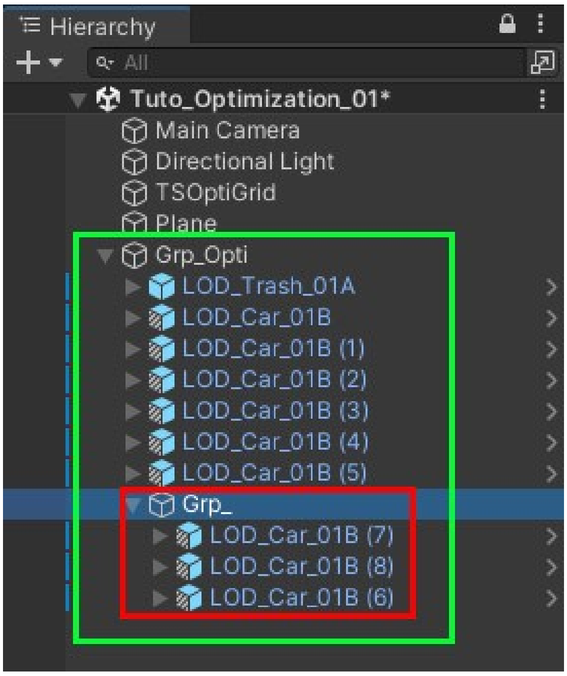
-If you want to hide Grid gizmo unckeck Show Gizmo checkbox (spot 3)



Important:
Each element included in Grp_Opti (Green) is considered as one object.
If a grp containing many objects(Red) is included into Grp_Opti (Green) it will be considered as one object.

<u>For good optimization:</u>
In the Grp_Opti(Green) avoid putting groups (Red) containing too many objects.

This also applies to distance groups (case 2)



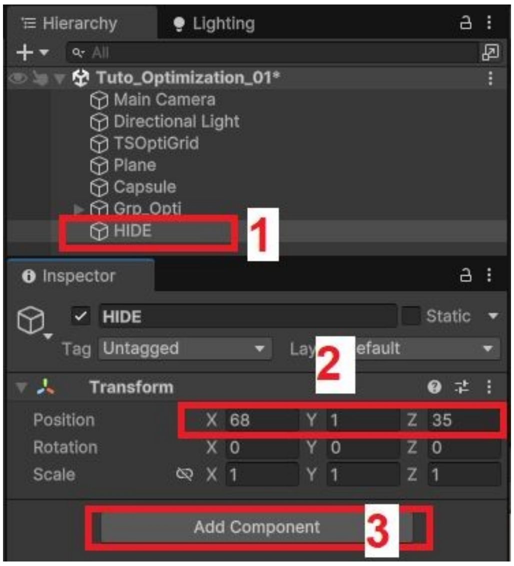
Case 2: Group Distance

-In hierarchy tab:
 create a new empty object(spot 1)

-Rename it HIDE for example.

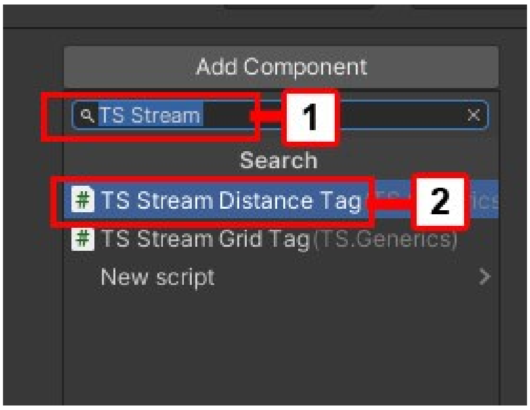
- For this example move the object to position (spot 2):
X = 68 | Y = 1 | Z = 35

-In Inspector tab press Add component button (spot 3)



-In search field type TS Stream (spot 1)

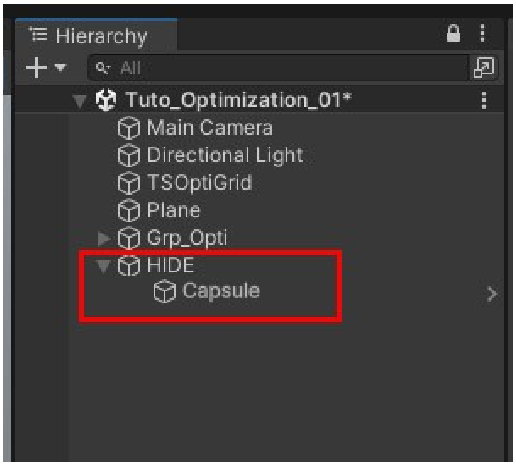
-Choose TS Stream Distance Tag (spot 2)



-Add Capsule to HIDE Group

-Set transform position of Capsule to
X =0
Y= 0
Z= 0

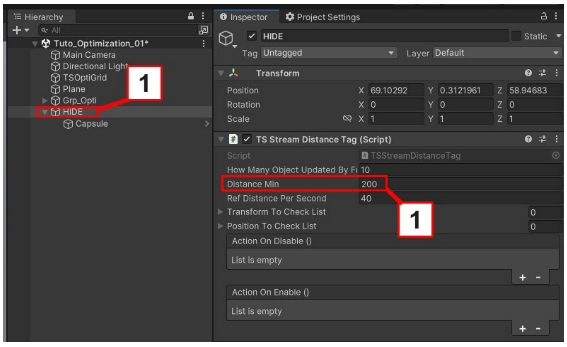
Capsule is now at the center of HIDE Group



Reminder: To determine the distance, the optimization system takes into account the center of the group

-Select HIDE Group (spot 1)

-Set Distance Min to 200 (spot 2)



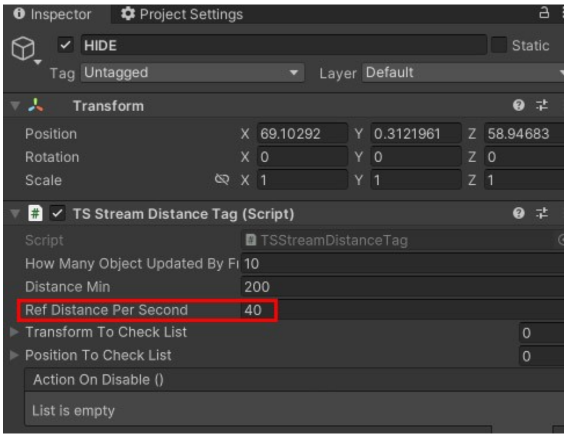
When the car player is within 200 meters of center of HIDE group objects are activated.
When the car player is more than 200 meters of the center of HIDE group, objects are deactivated.

Test Interval:

The script tests the distance between the player and the group at regular intervals.
Depending on the car player's movement speed, it is necessary to adjust the interval.

If the parameter is badly adjusted, the objects will appear too late (clipping) or too early (bad optimization)

To adjust the interval set Ref Distance Per second parameter as needed
Tips: If you find that objects appear too late (clipping) increase the value



Grid optimization (coding)

<u>If you want initialize manually the OptimizationGrid system from any script call</u>:

```
HP.Generics.TSOptiGrid.instance.Init();
```

<u>To update manually the OptimizationGrid System, from any script :</u> call these 2 lines inside a coroutine:

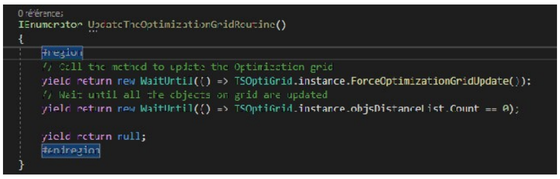
(It is useful if you want to create a spawn system for example)

```
// Call the method to update the Optimization grid

yield return new WaitUntil(() =>
HP.Generics.TSOptiGrid.instance.ForceOptimization
GridUpdate());

// Wait until all the objects on grid are updated

yield return new WaitUntil(() =>
HP.Generics.TSOptiGrid.instance.objsDistanceList.
Count == 0);
```



Delete Optimization

To delete optimization system in your scene:
In hierarchy tab delete TSOptiGrid

Troubleshooting

Console Warning or Error :
The type or namespace name universal does not exist

This means that URP Package is not installed.

If you do not have URP installed, other error messages may appear in the console. Install URP to make console errors disappear.

All shaders are pink

This means that URP Package is not installed or configured incorrectly.

Install URP to solve the problem

The textures of roads does not integrate well with the textures of the ground

You must calculate lightmaps.

To learn how to calculate lightmaps Link

Lods prefabs disappear too close to the camera

In Project settings > Quality
-Set Lod Bias to 2

During the game, some objects do not appear or appear / disappear at the wrong moment

Try these different solutions:

In Project settings > Quality
-Set Lod Bias to 2.

-To determine the distance, the optimization system (script included in the asset) takes into account the center of the group. Check that the center of the group.
-Set up GridOptimization row and columns.
-Set up Distance Min.
-Set up Ref Distance Per second. For more informations about optimization system Link

I can't draw the points to create a road

To solve this problem:
-Verify that gizmos are activate in scene tab
-In scene view select the terrain
-In hierarchy tab select again the new Road
-Press keyboard shortcut N

For more informations Link

Console Warning :
There are x objects in the scene with overlapping Uv's

Do not take this warning into account: it has no impact on rendering quality or performance.

There are invisible colliders on terrain:

After creating roads you must delete roads colliders

For more informations Link

Console Warning :
InitializeLighmapData job with hash .... exit code 2

This means that there is not enough RAM to calculate the lightmaps

To solve the problem:

-Reduce the number of prefabs into the scene